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RESEARCH ARTICLE OPEN ACCESS

Bias Correction of ERA5 Temperature Across Five Topoclimatic Clusters: A Comparative Evaluation of Four Methods

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Received: 8 January 2026 | **Revised:** 17 May 2026 | **Accepted:** 19 June 2026

Keywords: ERA5 | machine learning | South America | temperature bias correction | topoclimatic clustering

ABSTRACT

Near-surface air temperature from the ERA5 reanalysis is widely used for climate analysis, impact studies and environmental modelling over South America, but it exhibits systematic and regionally dependent biases that limit its direct applicability. This study aims to improve ERA5 daily maximum (TMAX) and minimum (TMIN) temperatures through physically informed regionalisation and bias correction. South America was clustered into homogeneous topoclimatic regions using *k*-means, spectral clustering and Gaussian Mixture Models, based on ERA5 thermodynamic variables, geographic coordinates and elevation, with the latter providing the most physically consistent partition into five clusters. Within each cluster, four bias-correction approaches were evaluated: Mean Bias (MB) and Model Output Statistics (MOS), applied at the cluster level, and Spatial Bias Correction (SBC) and a Multi-Layer Perceptron (MLP), which account for spatial variability across the broader domain. All methods were calibrated and assessed using independent subsets of in situ observations. ERA5 shows a pervasive cold bias in TMAX and a warm bias in TMIN, with the largest errors occurring in mountainous and high-latitude regions. All methods reduce these errors, with the MLP providing the most consistent improvements across regions and seasons. However, its performance is interpreted in light of its greater flexibility and ability to represent non-linear spatial–temporal relationships using physically meaningful predictors, rather than as evidence of universal superiority. The comparison reflects differences in modelling approaches, from linear and spatial interpolation to non-linear learning, within a consistent regionalised framework. Model performance is influenced by the spatial distribution of observations. In data-sparse regions such as the Amazon, MB, MOS and SBC are constrained by limited station coverage, whilst the MLP can partially compensate through cross-regional generalisation, which may reflect the transfer of relationships learned in data-rich regions. In mountainous and high-latitude regions, all methods are affected by observational limitations and unresolved physical processes, resulting in higher uncertainty. Overall, the MLP demonstrates strong effectiveness for bias correction and shows potential for predictive applications.

1 | Introduction

Near-surface air temperature is a fundamental variable for climate monitoring, impact assessment and numerical weather and climate modelling. Over South America, temperature exerts strong control on key environmental processes, including evapotranspiration (Fleischmann et al. 2023), ecosystem productivity, hydrological extremes (De Araújo et al. 2022) and energy demand, whilst also shaping human vulnerability to heatwaves (Feron et al. 2019), cold spells and long-term climate change. The continent exhibits pronounced thermal heterogeneity, driven by its wide latitudinal extent, complex topography,

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sharp land–sea contrasts and diverse atmospheric circulation regimes (Junquas et al. 2024). Accurately representing daily maximum (TMAX) and minimum (TMIN) temperatures across such a heterogeneous domain therefore remains a major challenge for both numerical models and reanalysis products (Ramarao et al. 2024).

Global atmospheric reanalyses, particularly ERA5 produced by the European Centre for Medium-Range Weather Forecasts (ECMWF), have become indispensable tools for climate research and operational applications due to their long temporal coverage, spatial completeness and physical consistency (Hersbach et al. 2020). ERA5 provides hourly estimates of near-surface temperature at relatively high spatial resolution and assimilates a vast array of in situ and satellite observations. Despite these advances, numerous studies have shown that ERA5 exhibits systematic and regionally dependent biases in near-surface temperature, especially over regions characterised by complex terrain, strong land–atmosphere coupling or sparse observational coverage (Betts et al. 2019; Yu et al. 2022; Wolf et al. 2025). Over South America, these biases typically manifest as an underestimation of daytime temperatures and an overestimation of nocturnal temperatures, leading to a reduced diurnal temperature range and seasonally varying errors (Rozante et al. 2022).

These deficiencies are closely linked to limitations in the representation of surface energy balance (Martens et al. 2020), cloud–radiation interactions (Moher et al. 2023), and boundary-layer processes (Dias-Júnior et al. 2022), as well as to unresolved subgrid topographic variability (Wang et al. 2025). In mountainous regions such as the Andes and southern Patagonia, coarse spatial resolution hampers the accurate depiction of elevation-dependent lapse rates (Praskievicz 2018; Dutra et al. 2020), snow–albedo feedbacks (Guo and Yang 2022), and local circulations. In tropical and subtropical lowlands, errors in cloud cover (Dommo et al. 2022), soil moisture (Shi et al. 2025), and turbulent mixing (Wei et al. 2025) affect both daytime heating and nocturnal cooling. As a consequence, direct application of ERA5 temperature data in impact studies, model evaluation, or downscaling frameworks may lead to systematic distortions if these biases are not properly addressed (Ascenso et al. 2024; Feng et al. 2025).

Statistical bias correction has therefore become a standard practice to improve the usability of reanalysis temperature fields. Traditional approaches, including simple mean bias adjustment and Model Output Statistics (MOS) (Glahn and Lowry 1972), are computationally efficient and easy to interpret, but they rely on linear assumptions and often fail to capture non-linear interactions between temperature, topography, seasonality, and land–atmosphere processes (Vrac and Friederichs 2015). More advanced spatial techniques, such as spatial bias correction based on geostatistical interpolation, partially address spatial heterogeneity but remain limited in their ability to represent complex, regime-dependent error structures (Dhawan et al. 2024).

In recent years, machine learning methods have emerged as powerful alternatives for bias correction, owing to their capacity to model non-linear relationships and high-dimensional dependencies (Seo and Ahn 2023; Nivron et al. 2024; Mauceri et al. 2025). Neural-network-based approaches, in particular Multi-Layer Perceptrons (MLPs), have shown promising results in correcting temperature biases by integrating spatial, temporal and meteorological predictors (Xu et al. 2021; Cho et al. 2020; Rozante et al. 2023). In this context, Li et al. (2020) demonstrated that data-driven models can effectively capture complex relationships between model outputs and observations, outperforming traditional statistical approaches in reducing systematic temperature biases. However, their performance is strongly influenced by the underlying climatic heterogeneity of the domain and by the availability of representative training data (Mfetoum et al. 2024; Surana et al. 2024; Yu et al. 2025). Applying a single correction model over climatically diverse regions such as South America may therefore mask important regional differences and limit generalisation. An effective strategy to address this issue is to combine bias correction with climate regionalisation (Tanimu et al. 2024; Le et al. 2024). By partitioning the study domain into homogeneous topoclimatic regions, it becomes possible to calibrate correction models that are physically consistent with local climatic regimes (Gómez-Navarro et al. 2018; Al-Rawas et al. 2025). Clustering techniques based on thermodynamic variables, geographic coordinates and elevation provide an objective framework for identifying such regions and have the potential to enhance both statistical and machine-learning-based corrections (Carbone et al. 2024; Lehner et al. 2025).

Against this backdrop, the primary objective of this study is to develop and evaluate a robust, regionally tailored framework for correcting systematic biases in ERA5 TMAX and TMIN across South America. This aim is pursued through three specific, interlinked goals: (i) to objectively partition the continent into homogeneous topoclimatic regions using a clustering methodology that optimally integrates thermodynamic and geographic drivers; (ii) to conduct, within each defined region, a comprehensive comparative evaluation of four distinct bias-correction approaches (Mean Bias adjustment (MB), Model Output Statistics (MOS), Spatial Bias Correction (SBC), and a Multi-Layer Perceptron (MLP) neural network); and (iii) to rigorously quantify the resulting improvements in temperature estimates using independent in situ observations, thereby identifying the most effective correction strategy for each major climatic regime. The structure of the paper reflects this progression: Section 2 details the study area, observational and reanalysis datasets, the topoclimatic clustering procedure, and the implementation of each bias-correction method. Section 3 presents the results, organised by topoclimatic cluster, assessing the seasonal performance of each technique. Finally, Section 4 discusses the physical drivers of the observed biases, the relative merits and limitations of the correction methods, and the broader implications of the findings, followed by concluding remarks in Section 5.

2 | Data and Methodology

2.1 | Area of Study

South America (Figure 1a) is the fourth largest continuous landmass on Earth (Alvarez-Varas et al. 2016), extending from about 12° N to 55° S, between the Gulf of Darién and the Tierra del Fuego archipelago (Figure 1). It is bordered by the Caribbean Sea, the Atlantic Ocean, and the Pacific

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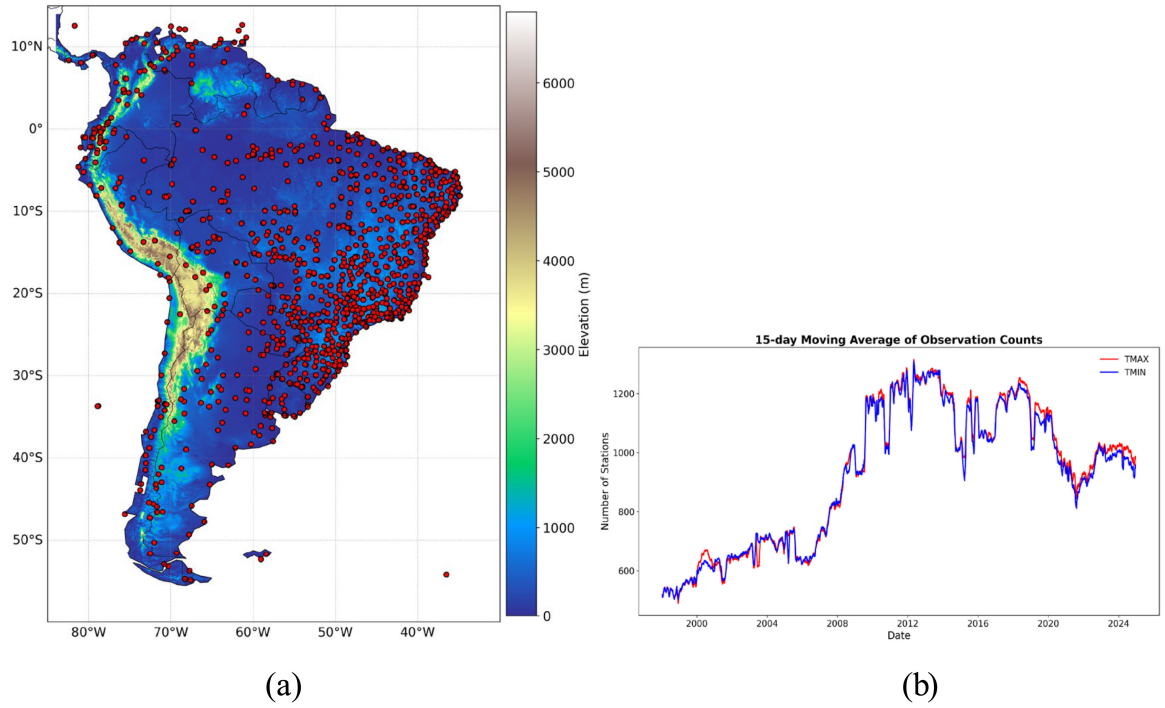


FIGURE 1 | (a) Spatial distribution of meteorological stations across South America, shown as red dots over the GTOPO30 topography. (b) Temporal evolution of the number of stations reporting TMAX (red) and TMIN (blue) temperatures, represented as 15-day moving averages. [Colour figure can be viewed at wileyonlinelibrary.com]

Ocean, which together exert major control on its atmospheric and oceanic circulation (Satyamurty et al. 1998). The Andes Mountains, stretching for nearly 7000 km along the western margin, form the longest and second-highest mountain chain globally and play a key role in modulating air flow and the transport of Amazonian moisture towards subtropical regions (Vera et al. 2006; Insel et al. 2010). Strong thermal contrasts arise from the continent's broad latitudinal and altitudinal range: equatorial regions such as the Amazon Basin show mean annual temperatures above 26°C, whilst subtropical and temperate zones range between 14°C and 22°C. In contrast, the southern Andes and Patagonia exhibit averages below 10°C, and the Atacama Desert in northern Chile remains one of the driest and coldest environments on Earth ($< 2\,\text{mm year}^{-1}$; Ritter et al. 2019). Overall, the continent can be divided into tropical, temperate, dry, and cold climatic zones (Garreaud et al. 2009), shaped by topography, latitude, and oceanic influences. Rainfall is abundant across most of South America, except in arid and semi-arid areas such as coastal Peru, northern Chile, Patagonia, and north-eastern Brazil.

2.2 | Observed Data

In this study, observational data comprising daily maximum temperature (Tmax), daily minimum temperature (Tmin), and altitude (H) were employed. These datasets originate from various meteorological station networks routinely processed and disseminated by the National Centre for Weather and Climate Research of the National Institute for Space Research (CPTEC/INPE). Regarding temporal resolution, some sources provide daily reports, such as the Surface Synoptic Observations (SYNOP) integrated into the Global Telecommunication System (GTS), from which TMAX and TMIN values are directly obtained. Other sources, including Aerodrome Meteorological Reports (METAR), Automatic Data Collection Platforms (PCDs), and regional meteorological centres, supply hourly temperature time series, from which daily TMAX and TMIN are derived based on the diurnal temperature cycle recorded at each station.

Meteorological data, even when representing observed conditions, may contain systematic or random errors originating from uncalibrated or ageing sensors, acquisition faults, or transmission failures. To ensure data reliability, a two-step operational quality control procedure developed by CPTEC/INPE was applied. The objective stage involves automated procedures for outlier detection (Tukey 1977), internal consistency verification (Baker 1992), and spatio-temporal validation (Xu et al. 2014), followed by a subjective stage in which a meteorologist reviews flagged records and determines whether to retain or discard them.

After quality control, the spatial distribution of meteorological stations across South America (Figure 1a) reveals a clear heterogeneity. The highest station density occurs in southeastern Brazil, particularly along the coastal and interior regions characterised by higher population density and well-developed infrastructure. Moderate densities are also found in northern Argentina, Uruguay, and central Chile. In contrast, station coverage is markedly sparse across the Amazon Basin, the Andean highlands, and southern Patagonia. These gaps reflect the combined effects of environmental constraints and logistical challenges associated with maintaining observation networks in remote or topographically complex areas. Overall, the network shows a bias towards low- and mid-elevation regions, whereas

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high-altitude and tropical rainforest environments remain underrepresented.

In addition to this spatial variability, the temporal evolution of the number of temperature observations across South America provides further insight into the development and consistency of the monitoring network. Based on the INPE quality-controlled dataset (Figure 1b), the curves represent the 15-day moving averages of the number of stations reporting TMAX (red) and TMIN (blue). The record shows a consistent long-term increase in the number of active stations from the late 1990s until approximately 2011, reflecting the expansion and consolidation of regional observational networks. Between 2010 and 2019, the number of available observations reached its maximum, with over 1200 stations reporting regularly, providing the most comprehensive and spatially representative coverage within the entire period. This decade therefore offers an optimal temporal window for climate analysis, ensuring both spatial consistency and temporal continuity in the dataset. After 2019, a gradual decline and higher variability are observed, possibly associated with interruptions in data transmission or maintenance issues affecting automatic stations. Overall, the 2010–2019 period represents the most stable and data-rich phase of the INPE observation network, forming a robust foundation for the subsequent analyses presented in this study.

2.3 | ERA5 Reanalysis

The ERA5 reanalysis, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), represents the fifth generation of global reanalyses and supersedes ERA-Interim, offering substantial improvements in spatial and temporal resolution as well as in the representation of atmospheric processes (Hersbach et al. 2020). From this dataset, hourly air temperature and 2-m dew point temperature were utilised, with daily maximum and minimum temperatures derived from the diurnal cycle. ERA5 provides hourly estimates of a wide range of atmospheric, land-surface, and oceanic variables at a horizontal resolution of approximately 31 km and 137 vertical levels, using a four-dimensional variational (4D-Var) data assimilation system that integrates model forecasts with in situ and satellite observations. It also includes a 10-member ensemble of data assimilation (EDA) to estimate uncertainty, available at ≈63 km resolution and 3-hourly frequency. The dataset extends from 1940 to the present and is updated daily with a latency of about 5 days. All data are freely accessible via the Copernicus Climate Data Store (https://cds.climate.copernicus.e).

2.4 | Topographic Data

The Global 30 Arc-Second Elevation Dataset (GTOPO30) (Earth Resources Observation and Science (EROS) Center 2017) was employed as the source of topographic information for this study, restricted to the spatial domain encompassing the South American continent. GTOPO30 is a global digital elevation model (DEM) with a horizontal resolution of 30 arc sec, equivalent to approximately 1 km at the equator. Developed under the coordination of the Earth Resources Observation and Science (EROS) Center of the United States Geological Survey (USGS), the dataset was produced through the integration of multiple raster and vector topographic data sources. It provides terrestrial elevation data only, excluding bathymetric information, and was generated through an international collaboration involving NASA, UNEP/GRID, USAID, INEGI (Mexico), GSI (Japan), Manaaki Whenua Landcare Research (New Zealand), and the Scientific Committee on Antarctic Research (SCAR).

Based on this dataset, the spatial pattern of topography across South America (Figure 1a) reveals a marked contrast between the continent's major physiographic regions. The Andes mountain chain dominates the western margin, extending from northern Colombia to southern Chile and Argentina, with elevations exceeding 6000 m in the Central Andes. Eastwards, elevation decreases sharply towards the extensive lowlands of the Amazon Basin, the Gran Chaco, and the Brazilian Highlands. The Brazilian Plateau presents moderate elevations, generally between 500 and 1500 m, whilst the Amazon and La Plata basins are predominantly low-lying areas below 200 m.

2.5 | Topoclimatic Clustering

To evaluate bias-correction methods, a cluster analysis was performed with the aim of delineating regions exhibiting similar thermal characteristics. The analysis incorporated monthly mean air temperature and monthly mean dew point temperature from the ERA5 reanalysis, as well as latitude, longitude, and elevation, the latter obtained from the GTOPO30 digital elevation model. The topographic data were interpolated to match the spatial resolution of the ERA5 temperature fields, ensuring consistency across all input variables. This approach allowed the identification of coherent regional clusters representing areas with homogeneous thermal and geographic patterns, facilitating a more robust application of statistical bias-correction techniques. By combining thermodynamic variables with topography, the clustering captures realistic spatial gradients of temperature influenced not only by altitude and latitude, but also by proximity to the ocean, which is known to modulate local and regional climatic conditions.

The primary purpose of the clustering framework is to provide a climatically stratified basis for the evaluation of the bias-correction methods, allowing their performance to be analysed separately across regions with broadly similar thermal and topoclimatic characteristics. This approach aims to avoid mixing fundamentally different climatic environments within a single statistical assessment over the highly heterogeneous South American domain. Although several physical mechanisms associated with ERA5 temperature biases are themselves climate-dependent, the methodology does not require the assumption that the identified clusters correspond to spatially homogeneous bias regimes.

To ensure a comprehensive assessment of clustering performance, three widely used methods were tested: *k*-means (Sreekala and Babu 2025), spectral clustering (Ng et al. 2001), and the Gaussian Mixture Model (GMM) (Ojha 2019; Yu and Deng 2015). These techniques were chosen because they represent distinct conceptual approaches, namely partition-based, graph-based, and probabilistic clustering. This

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diversity allowed a robust evaluation of how effectively each method captures thermal and topographic gradients across South America.

The evaluation of clustering performance was based on quantitative metrics and a systematic exploration of the number of clusters. For this purpose, clustering solutions were computed considering values of k ranging from 2 to 15 for all methods. The silhouette coefficient (Rousseeuw 1987) was used to assess clustering performance, providing a measure of cluster compactness and separation. Additionally, the Bayesian Information Criterion (BIC) (Webster 2020) was employed to determine the optimal number of clusters, providing a balance between model fit and complexity. As BIC is inherently defined for probabilistic models, it was applied within the Gaussian Mixture Model framework, following the identification of the best-performing method based on the silhouette analysis.

Figure 2 summarises the results of this evaluation. Panel (a) presents the variation of the silhouette coefficient for the

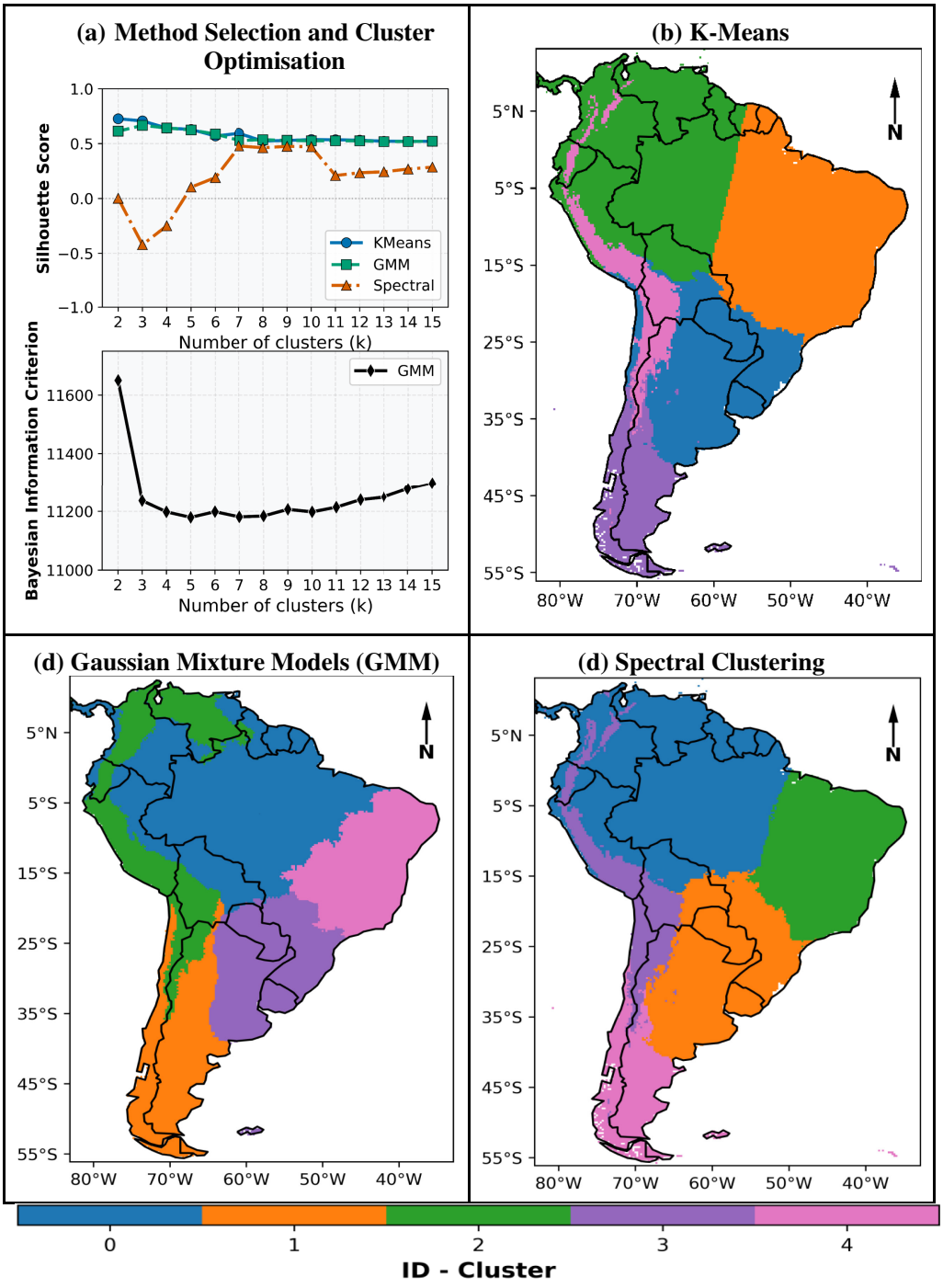


FIGURE 2 | (a) Method Selection and Cluster Optimisation, and spatial distribution of topoclimatic clusters across South America derived from three clustering methods: (b) k-means, (c) Gaussian Mixture Model (GMM), and (d) Spectral Clustering. Each colour represents a distinct cluster. [Colour figure can be viewed at wileyonlinelibrary.com]

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different clustering methods, together with the BIC values for the GMM across the tested range of k. The results indicate that k-means and GMM exhibited very similar performance in terms of silhouette scores, both outperforming spectral clustering. The BIC curve displayed a clear minimum at $k = 5$, indicating the most parsimonious and statistically robust solution. Based on these results, $k = 5$ was selected as the optimal number of clusters. Panels (b)–(d) of Figure 2 present the spatial distribution of clusters obtained using k-means, GMM, and spectral clustering, respectively. Although all methods reproduced the broad climatic divisions of the continent, their ability to represent transitional and topographically modulated zones varied significantly. The k-means method produced more abrupt and spatially homogeneous boundaries, particularly over central Brazil and the Andean region, indicating limited sensitivity to gradual temperature gradients. Spectral clustering showed some improvements in specific areas but remained less effective in capturing complex altitudinal transitions, particularly over Venezuela.

Amongst the evaluated methods, the GMM produced the most coherent and physically consistent representation of climatic patterns. Although its statistical performance was comparable to that of k-means, its probabilistic formulation assigns each grid point a likelihood of belonging to different clusters, thereby enabling smooth transitions between regions. This characteristic is essential for environmental variables such as temperature, which vary continuously in space. The GMM more effectively captured contrasts associated with altitude, distance from the ocean, and thermodynamic gradients, resulting in smoother and more realistic zonation. From a climatological perspective, this behaviour is particularly relevant in South America, where complex interactions between topography and atmospheric processes lead to gradual spatial transitions rather than abrupt boundaries. For these reasons, the GMM was selected as the most suitable method for defining topoclimatic regions in this study, combining robust statistical support with a physically meaningful representation of climatic variability.

2.6 | Bias-Correction Methods

This subsection describes the four bias-correction methods evaluated in this study, outlining their main characteristics and implementation. The selected approaches were chosen to provide a structured comparison across different levels of complexity. Mean Bias (MB) and Model Output Statistics (MOS) represent simple and widely used linear methods, whilst Spatial Bias Correction (SBC) explicitly accounts for spatial variability, which is particularly relevant over South America given its complex topography and uneven station distribution. The Multi-Layer Perceptron (MLP) represents a non-linear, data-driven approach capable of modelling interactions between temperature, seasonality, and geographic factors.

More advanced distribution-based methods, such as (QDM) and scaled distribution mapping (SDM) (Li and Li 2023), were not considered because the focus is on improving the accuracy and spatial consistency of daily temperature fields within topoclimatic regions, rather than adjusting the full statistical distribution. In addition, the selected methods enable a more controlled comparison between linear, spatial, and non-linear approaches within the clustering framework.

It should also be noted that the statistical methods (MB, MOS, SBC) assume stationarity in the relationship between ERA5 and observations, which may limit their applicability under changing climate conditions. Therefore, the results are primarily representative of present-day climate, and the application of non-stationary or distribution-based approaches remains a subject for future work. In this context, the following subsections detail the formulation and implementation of each method:

a. *Model Output Statistics (MOS)*: constitute a long-standing and widely adopted statistical framework for correcting systematic biases in gridded atmospheric datasets (Glahn and Lowry 1972). For several decades, MOS techniques have been applied to improve the representation of essential meteorological variables, including air temperature, precipitation, and wind speed, by statistically aligning modelled (Yang et al. 2020) or reanalysis fields (Tahir et al. 2021) with ground-based or remotely sensed observations (Verbois et al. 2023). In the context of bias correction for reanalysis products, a particularly common application is the adjustment of ERA5 fields through statistical relationships derived from observed measurements (denoted here as OBS). The fundamental MOS formulation relies on ordinary least squares (OLS) linear regression, in which OBS serves as the predictand and ERA5 as the predictor. The mathematical formulation used to correct the ERA5 series is presented below. The correction is obtained by applying a linear transformation fitted between the observations (OBS) and the original ERA5 values, expressed as:

$$\mathrm{ERA}^{\mathrm{corr}}_{(i,j)} = a + b * \mathrm{ERA}_{(i,j)} \tag{1}$$

The coefficients a (additive bias) and b (multiplicative bias) are estimated using ordinary least squares (OLS) across the n stations, with indices i, j indicating the latitude and longitude of the corresponding grid point.

$$b = \frac{\sum_{k=1}^{n} \left(\mathrm{ERA5}_k - \underline{\mathrm{ERA5}}\right) * \left(\mathrm{OBS}_k - \underline{\mathrm{OBS}}\right)}{\sum_{k=1}^{n} \left(\mathrm{ERA5}_k - \underline{\mathrm{ERA5}}\right)^2} \tag{2}$$

$$a = \underline{\mathrm{OBS}} - b * \underline{\mathrm{ERA5}} \tag{3}$$

Despite its robustness and interpretability, MOS relies on a linear relationship between ERA5 and observations and assumes temporal stationarity. As a result, it may not fully capture non-linear processes or regime-dependent biases, particularly in regions with strong land–atmosphere coupling or complex topography, potentially leading to residual errors under certain conditions.

b. *Mean Bias (MB)*: is one of the simplest and most commonly (Kanda and Fletcher 2024; Staehle et al. 2024) applied techniques for reducing systematic errors in environmental and climate datasets. It assumes that the primary discrepancy between a modelled or gridded dataset and ground-based observations can be represented by a constant additive offset. This offset is estimated as the average difference between ERA5 temperatures and the corresponding observations (OBS), allowing the

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systematic bias to be removed through a straightforward adjustment.

$$\mathrm{MB} = \frac{1}{n}\sum_{k=1}^{n}\left(\mathrm{ERA5}_k - \mathrm{OBS}_k\right) \tag{4}$$

$$\mathrm{ERA5}_{(i,j)}^{\mathrm{corr}} = \mathrm{ERA5}_{(i,j)} - \mathrm{MB} \tag{5}$$

However, MB assumes a constant additive bias and does not account for variability in space, time, or across temperature ranges. This simplification may limit its ability to correct seasonally varying or non-linear biases, often resulting in undercorrection or overcorrection in regions with heterogeneous climatic conditions.

c. *Spatial Bias Correction (SBC)*: For each observational station, the bias was first estimated as the difference between the ERA5 reanalysis at the nearest grid point and the corresponding station observation (ERA5 − OBS). These pointwise bias values were then spatially interpolated using ordinary kriging in order to produce a continuous daily bias field. The correction was subsequently applied by subtracting this interpolated bias field from the original ERA5 data, yielding a spatially adjusted product. This procedure was carried out independently for each day within the study period, ensuring that the temporal variability of the bias was fully accounted for. The mathematical formulation is described below.

i. Station-wise bias:

$$\mathrm{BIAS}_k = \mathrm{ERA5}_{ik,jk} - \mathrm{OBS}_k \tag{6}$$

ii. *Kriged bias field at grid cell (i,j): The* interpolated bias at grid cell (i,j) is a weighted sum of station biases:

$$\mathrm{BIAS}_{(i,j)} = \sum_{k=1}^{N} w_k(i,j).\,\mathrm{BIAS}_k \tag{7}$$

with

$$\sum_{k=1}^{N} w_k(i,j) = 1 \tag{8}$$

iii. *Bias correction at grid cell (i,j)*: Subtract the interpolated bias from the raw ERA5 grid value:

$$\mathrm{ERA5}_{(i,j)}^{\mathrm{corr}} = \mathrm{ERA5}_{(i,j)} - \mathrm{BIAS}_{(i,j)} \tag{9}$$

where

- $\mathrm{ERA5}_{ik,jk}$ be the grid point nearest to station k,
- OBS_k be the observation at station k,
- BIAS_k the station-wise bias at station k,
- $\mathrm{BIAS}_{(i,j)}$ the interpolated (kriged) bias at ERA5 grid cell (i,j),
- $w_k(i,j)$ the ordinary-kriging weights,
- $\mathrm{ERA5}_{(i,j)}^{\mathrm{corr}}$ the bias-corrected ERA5 at grid cell.

Although SBC accounts for spatial variability, its performance depends strongly on station density and the assumptions of the interpolation method. In data-sparse regions, such as the Amazon and the Andes, the interpolated bias fields may be less reliable, which can introduce spatial smoothing and reduce the accuracy of local corrections.

d. *Multi-Layer Perceptron (MLP)*: is a feed-forward artificial neural network composed of sequential layers of interconnected neurons capable of modelling complex non-linear relationships. Owing to its flexibility and ability to learn general patterns from data, the MLP is widely used in regression tasks and environmental applications, including meteorological bias correction. The input dataset comprised spatial descriptors (longitude (LON), latitude (LAT), and elevation (ELEV)), the ERA5 modelled temperature (ERA5) at each station location, and a suite of engineered temporal and meteorological predictors. These included cyclic representations of the day of year (DoY), expressed through sine and cosine transformations (DoY_SIN = sin(2π·DoY/365) and DoY_COS = cos(2π·DoY/365)), which preserves the periodic nature of the annual cycle. In addition, a coarse seasonal encoding was introduced using fixed-phase sinusoidal representations (SEASON_SIN = sin(ϕ) and SEASON_COS = cos(ϕ)), where ϕ denotes the phase associated with each climatological season along the annual cycle (summer: 0, autumn: $\pi/2$, winter: π, and spring: $3\pi/2$). This formulation enables the model to distinguish regime-dependent seasonal characteristics whilst maintaining the inherent cyclic structure of the calendar year. The target variable was defined as the temperature bias, calculated as the difference between ERA5 and the corresponding in situ observation (ERA5–OBS). The mathematical formulation corresponding to the variables and encodings described above is detailed below.

$$\widehat{\mathrm{BIAS}} = f_\phi(X) \tag{10}$$

where $\widehat{\mathrm{BIAS}}$ is estimated bias by the network, f_ϕ is MLP trained with parameters, X is input vector containing [LON, LAT, ERA5, DoY_SIN, DoY_COS, SEASON_SIN, SEASON_COS].

$$\mathrm{ERA5}_{(i,j)}^{\mathrm{corr}} = \mathrm{ERA5}_{(i,j)} - \widehat{\mathrm{BIAS}}_{(i,j)} \tag{11}$$

The architecture consisted of five hidden layers with 512, 256, 128, 64, and 32 neurons, respectively, followed by a single linear output neuron (Figure 3). This progressively narrowing, pyramidal structure was selected as it enables the network to capture high-dimensional spatial–temporal interactions in the initial layers, whilst the gradual reduction in width promotes increasingly abstract representations and helps control overfitting. Each hidden layer employed batch normalisation to stabilise activations, ELU activation to improve convergence and avoid vanishing gradients, dropout (0.3) to balance model capacity and regularisation, and progressively stronger L2 penalties to encourage more generalisable weight distributions. Training used the Adam optimiser with a conservative learning rate (5×10^{-5}) and mean squared error loss, with early stopping and adaptive learning-rate reduction based on validation RMSE to ensure stable and efficient convergence. Model generalisation was evaluated using a stratified train–validation split and five-fold cross-validation, and the best configuration was retained through checkpointing. Collectively, these design

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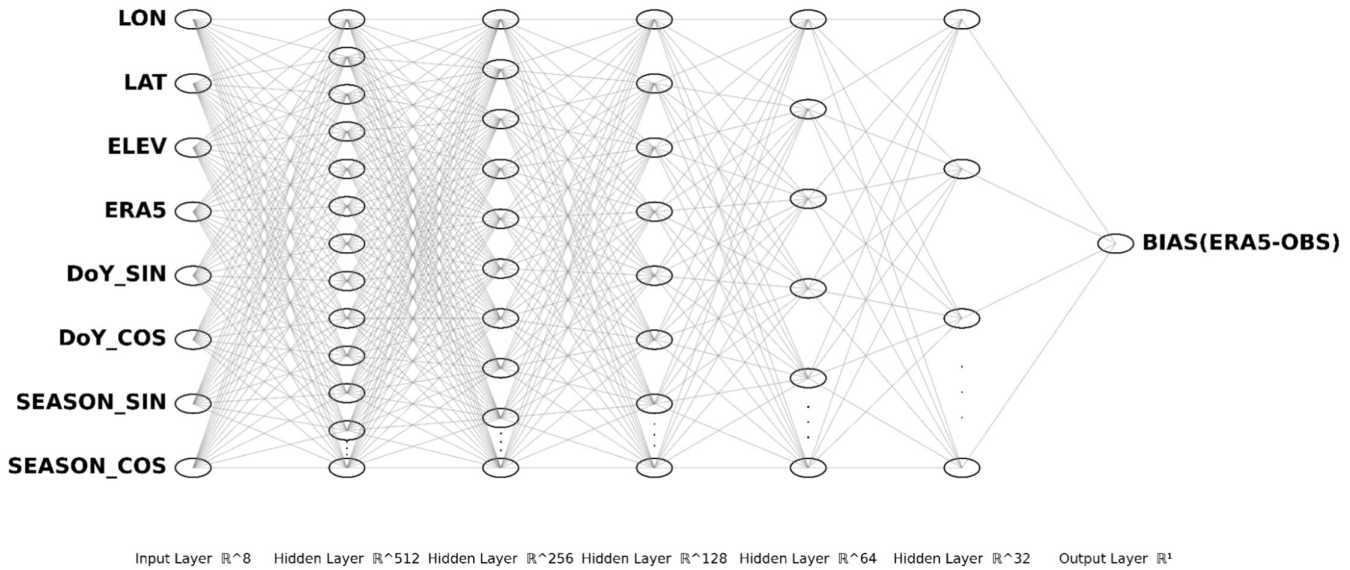


FIGURE 3 | Schematic representation of the Multi-Layer Perceptron architecture. The input layer includes spatial predictors (LON, LAT, ELEV), ERA5 modelled temperature, and engineered temporal predictors derived from the calendar date (DoY_SIN, DoY_COS, SEASON_SIN and SEASON_COS).

choices reflect a carefully balanced architecture aimed at maximising predictive performance whilst maintaining robustness across diverse meteorological conditions.

Despite its superior performance, the MLP approach is inherently data-driven and depends on the representativeness of the training dataset. It may be sensitive to overfitting and can exhibit reduced generalisation in regions with limited observations or under climate conditions not represented in the training period.

It is important to clarify the role of the topoclimatic clustering in the implementation of the bias-correction methods. Whilst the clusters are consistently used as a regional framework for evaluating the performance of all methods, they are explicitly incorporated into the calibration of only the MB and MOS approaches. In these methods, the bias correction is computed independently within each cluster, allowing the statistical relationships between ERA5 and observations to be conditioned on regionally homogeneous climatic regimes. This design is intended to better capture spatial variations in bias structure associated with topography and climate. In contrast, the SBC and MLP methods are applied over the full spatial domain without explicit use of the cluster partition during model calibration. The SBC method relies on spatial interpolation of station-based corrections, whereas the MLP is trained using the full training dataset spanning all clusters, allowing it to learn non-linear relationships across different climatic regimes. As a result, these methods do not explicitly rely on cluster-based regionalisation, but instead represent spatial variability through interpolation (SBC) or through implicit, data-driven regionalisation in feature space (MLP) to capture temperature biases.

2.7 | Evaluation Methods

The procedure adopted to evaluate the performance of each bias-correction method followed a sequence of steps, as summarised in Figure 4.

Step 1 *Climate regionalisation*: The study domain was first subdivided into five homogeneous climate regions using the Gaussian Mixture Model (GMM) clustering algorithm applied to the complete set of observational data (OBS_100%).

Step 2 *Selection of an independent evaluation dataset*: Within each climate region, 10% of the observations (OBS_10%) were randomly selected and withheld exclusively for model evaluation, ensuring that these data were not used during the calibration phase.

Step 3 *Definition of the calibration dataset*: The remaining 90% of the observations (OBS_90%) were used solely for the calibration of the bias-correction methods. This data partitioning constitutes a cross-validation framework, guaranteeing statistical independence between calibration and evaluation.

Step 4 *Calibration of bias-correction methods*: The bias-correction methods MOS, MB, SBC, and MLP were calibrated independently for each climate region using only the OBS_90% subset. This region-specific calibration allowed each method to account for the distinct climatic characteristics of each region.

Step 5 *Generation of bias-corrected ERA5 data and spatial matching*: Application of the calibrated methods yielded bias-corrected ERA5 fields for each technique. The independent OBS_10% observations were then spatially matched to the nearest ERA5 grid point of the corrected fields, forming paired observed–modelled datasets.

Step 6 *Statistical performance assessment*: Based on the paired bias-corrected ERA5 and OBS_10% data, standard performance metrics were computed, including the mean error (EM), mean absolute error (MAE), and the root mean square error (RMSE). These metrics enabled a robust comparison of the relative performance of the bias-correction methods across climate regions, assessing both accuracy and the ability to reproduce observed variability.

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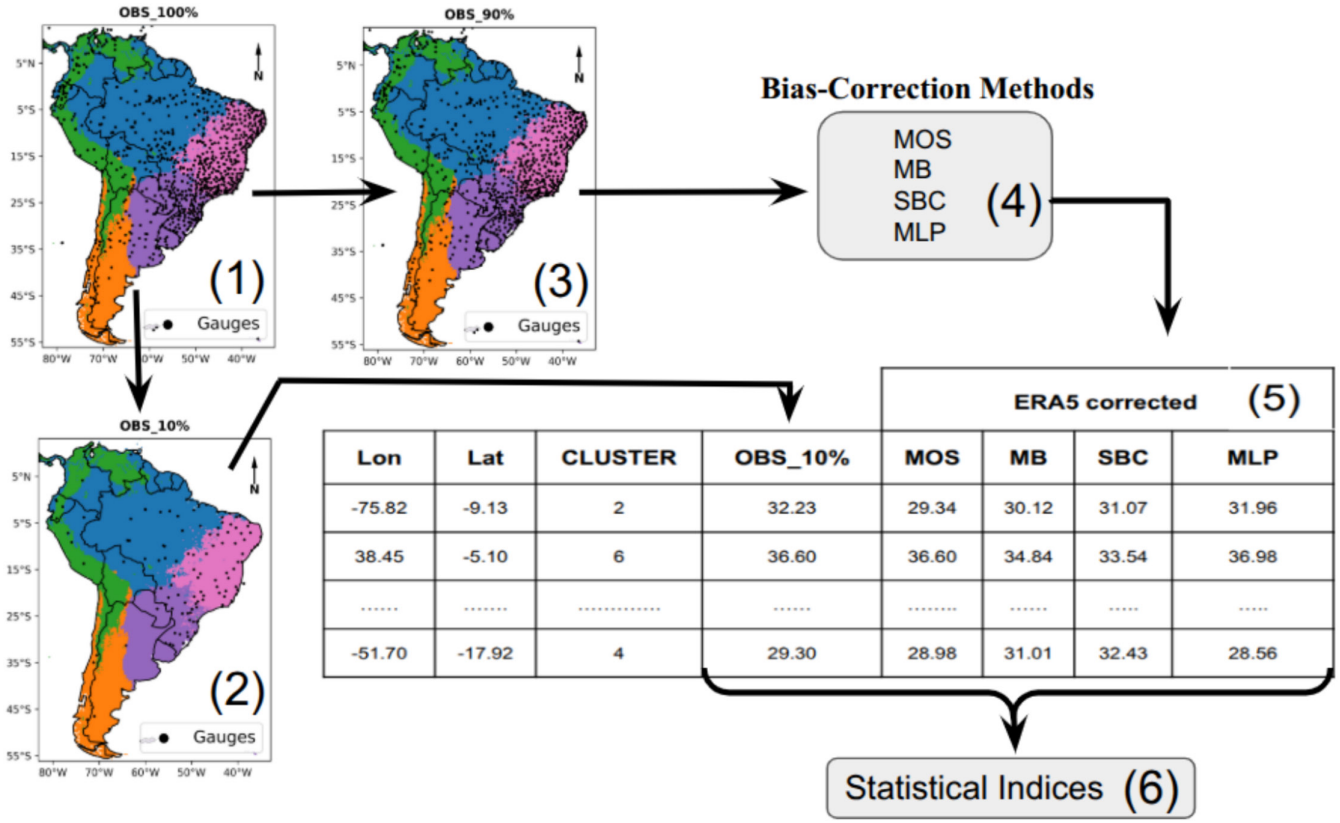


FIGURE 4 | Schematic representation of the evaluation framework adopted for the bias-correction methods. [Colour figure can be viewed at wileyonlinelibrary.com]

3 | Results

The bias-correction methods are applied at the daily scale, whereas the results are presented in terms of climatological trimesters, reflecting a deliberate and physically consistent methodological choice. Daily correction is required to properly address systematic errors associated with synoptic variability, diurnal temperature cycles, and day-to-day land–atmosphere interactions, which are particularly pronounced over South America due to its strong latitudinal gradients, complex topography, and heterogeneous surface conditions. Performing the correction at this temporal resolution ensures that high-frequency biases are effectively removed before any temporal aggregation. The subsequent trimestral presentation of results aims to emphasise robust climatic signals by reducing random variability and sampling noise, whilst aligning the analysis with the dominant seasonal modulation of temperature across the continent. This approach enhances statistical stability, facilitates inter-regional comparison, and allows a clearer interpretation of the performance of the bias-correction methods within distinct seasonal regimes, without compromising the physical consistency of the correction process.

3.1 | Cluster 0: Central-Northern Amazon Basin

Cluster 0 (Figure 2b shaded in blue) represents the dominant climatic regime over northern and central South America, encompassing most of the Amazon Basin, northern Brazil, parts of the Guianas, and extending westwards to the eastern slopes of the Andes. The region is characterised by persistently warm conditions, weak annual thermal amplitude, and limited topographic modulation, with near-surface temperatures strongly coupled to atmospheric humidity, cloud cover, and land–atmosphere interactions.

For TMAX (Figure 5a–c), the seasonal cycle is weak, with observed values increasing gradually from about 31°C in summer–autumn to slightly above 33°C in spring. ERA5 systematically underestimates TMAX throughout the year, exhibiting a cold bias of approximately 1.0°C–1.3°C, which may be associated with known difficulties in representing surface radiative processes under deep convection, extensive cloudiness, and strong evapotranspiration in humid tropical environments.

All bias-correction methods substantially reduce this underestimation, with the MLP nearly eliminating the systematic bias in all seasons. MOS, SBC, and MB retain small residual cold biases, particularly in summer and spring. Raw ERA5 shows the largest errors (MAE >1.6°C; RMSE ≈ 2.0°C–2.2°C), whilst bias correction markedly improves performance; the MLP yields the strongest reductions, lowering RMSE to about 1.3°C–1.5°C (≈30%–40% improvement), followed by MOS and SBC, with MB performing slightly worse.

For TMIN (Figure 5d–f), seasonal variability is even weaker, reflecting high humidity and frequent cloud cover that limit nocturnal radiative cooling. Observed TMIN decreases modestly from summer to winter (≈21°C) and increases again in spring. ERA5 exhibits a systematic warm bias in all seasons, ranging from about 0.4°C in summer–autumn to more than 1.0°C

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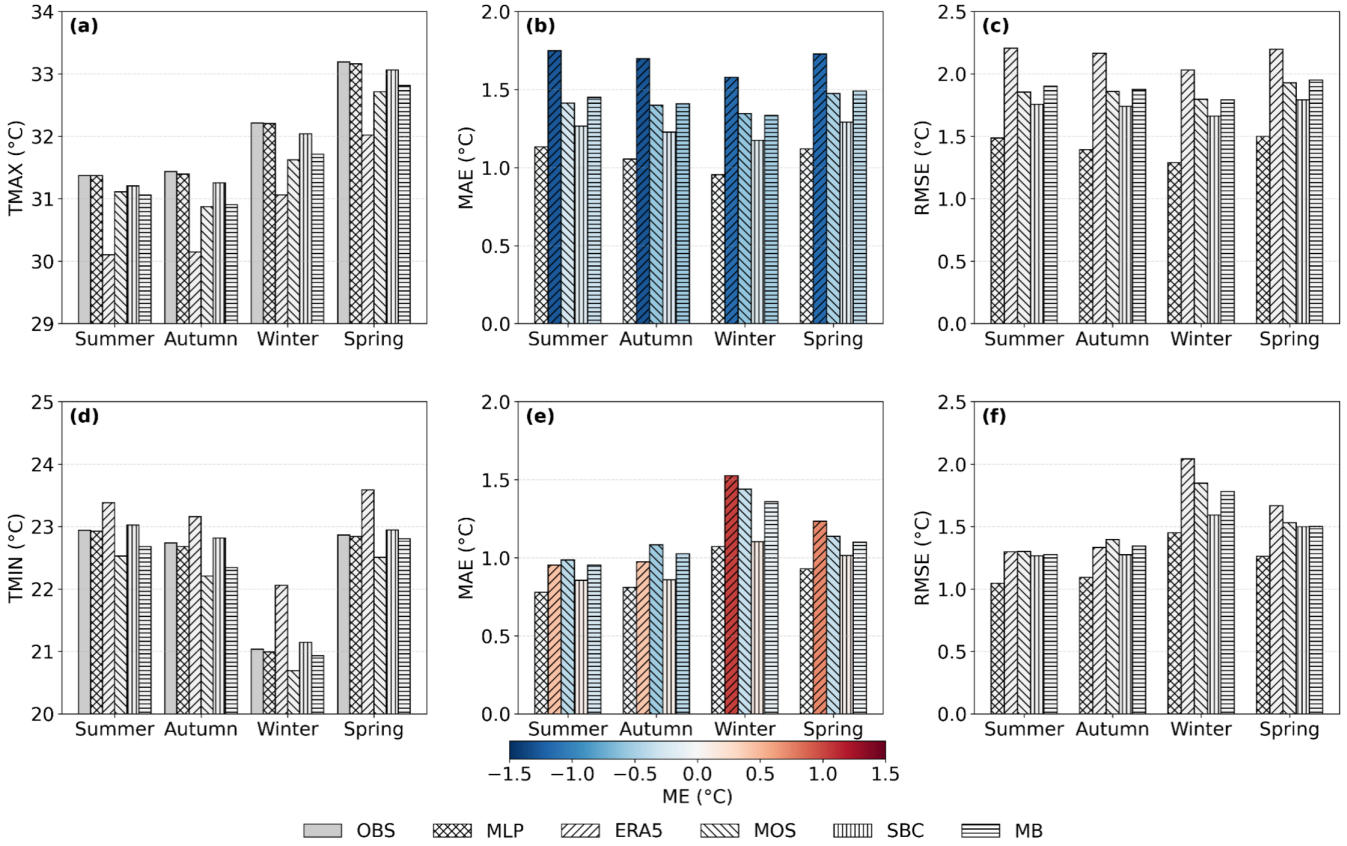


FIGURE 5 | Seasonal evaluation of MAX and TMIN for Cluster 0. Panels (a, d) show seasonal mean temperatures from observations (OBS), ERA5, and bias-corrected ERA5 using MOS, SBC, MB, and MLP. Panels (b, e) and (c, f) display the MAE and RMSE, respectively. Bar colours indicate the sign and magnitude of the ME. [Colour figure can be viewed at wileyonlinelibrary.com]

in winter, which may be related to difficulties in simulating nocturnal cooling and stable boundary-layer processes under humid tropical conditions. Bias-correction methods effectively reduce this overestimation, with the MLP again providing the best performance, achieving near-zero mean errors across seasons. MOS and MB slightly overcorrect in some cases, leading to small residual cold biases. Uncorrected ERA5 presents MAE values increasing from ~1.0°C in summer to >1.5°C in winter, with RMSE rising from ~1.3°C to just above 2.0°C; these errors are reduced by about 25%–30% with the MLP, which lowers RMSE to approximately 1.05°C–1.45°C.

Overall, TMAX exhibits consistently larger errors than TMIN in Cluster 0, indicating greater difficulty in representing daytime temperature extremes in this tropical regime. Although bias correction improves both variables, relative gains are more pronounced for TMIN, particularly in MAE, which is consistent with the stronger influence of boundary-layer processes on nocturnal temperatures and the more complex radiative and convective controls affecting TMAX.

3.2 | Cluster 1: Southern South America (Temperate and Patagonia Region)

Cluster 1 (Figure 2b shaded in orange) represents the cool-temperate to cold climatic regime of the southern Andes and Patagonia, extending from central–southern Chile into southern Argentina. Temperature patterns are primarily controlled by latitude, the strong influence of the Southern Ocean, and pronounced Andean orography. The region is characterised by low mean temperatures, cool summers and cold winters, with frequent mid-latitude frontal systems shaping seasonal variability. The Andes play a key role in modulating near-surface temperatures, favouring more maritime and thermally stable conditions on the windward Chilean side, whilst the leeward Argentine sector exhibits greater continentality and larger annual and diurnal temperature ranges.

For TMAX (Figure 6a–c), observed values decrease from summer to winter and increase again in spring. ERA5 systematically underestimates daytime temperatures throughout the year, with a cold bias exceeding 2°C in summer and spring, which may be associated with known limitations in representing surface energy balance, snow–albedo feedbacks, and boundary-layer processes in high-latitude and mountainous environments. All bias-correction methods substantially reduce this underestimation, with SBC and MLP nearly eliminating the systematic bias, whilst MOS and MB retain small residual errors. Uncorrected ERA5 exhibits large errors, with RMSE values close to or exceeding 3°C, whereas all correction approaches yield marked improvements; amongst them, the MLP consistently provides the lowest RMSE (around 2°C).

For TMIN (Figure 6d–f), a strong seasonal contrast is observed, with mean values decreasing from about 13.1°C in summer to 3.2°C in winter, followed by a rapid recovery in spring. ERA5 shows a systematic warm bias in all seasons (≈0.5°C–0.9°C), strongest during autumn and winter, which may be related

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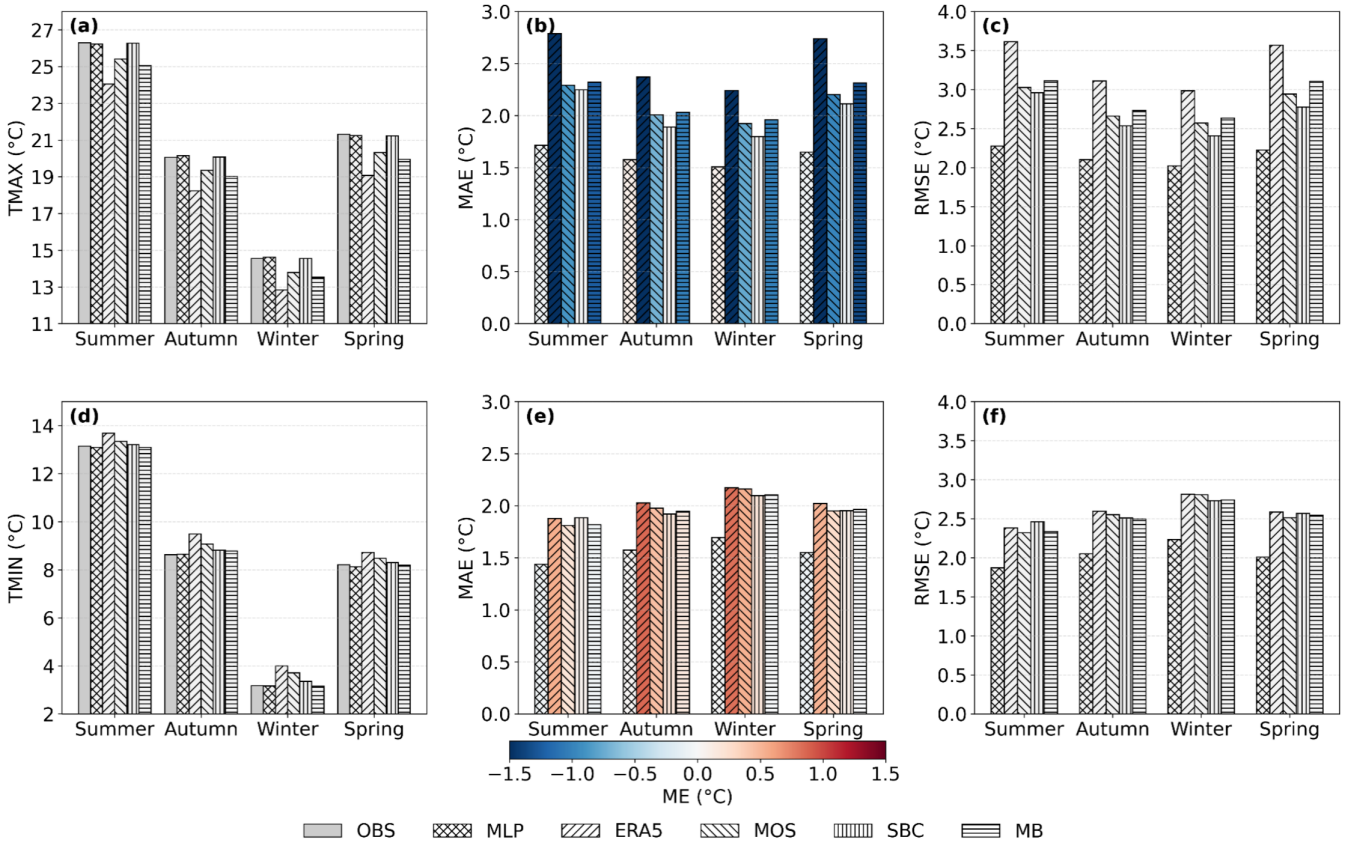


FIGURE 6 | Seasonal evaluation of MAX and TMIN for Cluster 1. Panels (a, d) show seasonal mean temperatures from observations (OBS), ERA5, and bias-corrected ERA5 using MOS, SBC, MB, and MLP. Panels (b, e) and (c, f) display the MAE and RMSE, respectively. Bar colours indicate the sign and magnitude of the ME. [Colour figure can be viewed at wileyonlinelibrary.com]

to difficulties in representing nocturnal cooling under stable boundary-layer conditions. Bias-correction methods effectively reduce this overestimation, with the MLP nearly removing the bias across all seasons, whilst MOS and SBC retain small warm biases during colder months. ERA5 MAE ranges from approximately 1.9°C to 2.2°C, with RMSE increasing from ~2.4°C in summer to nearly 2.8°C in winter; these errors are reduced by about 20%–25% with the MLP, which again achieves the lowest RMSE (≈1.9°C–2.2°C).

Overall, Cluster 1 exhibits a systematic cold bias in ERA5 TMAX and a warm bias in TMIN, leading to larger errors for daytime than for nocturnal temperatures. Bias-correction methods substantially improve both variables, with the MLP consistently outperforming the other approaches, yielding the lowest ME, MAE, and RMSE and demonstrating superior robustness in this cold, topographically complex region.

3.3 | Cluster 2: Western Tropical South America (Andes and Pacific Influence)

Cluster 2 (Figure 2b shaded in green) occupies a narrow, meridionally oriented belt along the Andes in western and north-western South America, extending from Colombia and western Venezuela through Ecuador, Peru and Bolivia to central–southern Chile and western Argentina, including the Chile–Argentina border region. The climate is strongly controlled by complex topography and high elevations, with terrain ranging from Andean foothills to high plateaus. As a result, temperatures are systematically cooler than in adjacent lowland regions, reflecting dominant altitude and lapse-rate effects. In tropical latitudes, annual mean temperatures are relatively stable but exhibit large diurnal ranges, whereas towards the subtropics seasonal temperature variability becomes more pronounced. The Andean chain also acts as a major climatic barrier, shaping atmospheric circulation, orographic precipitation and marked windward–leeward thermal contrasts.

Figure 7 summarises the seasonal behaviour of TMAX and TMIN and their associated errors for Cluster 2. For TMAX (Figure 7a–c), observed temperatures show limited seasonal variability, ranging from approximately 25.4°C in winter to 26.6°C in summer. ERA5 exhibits a strong and persistent cold bias in all seasons (ME ≈ −2.5°C to −2.8°C), with MAE values close to 3.0°C–3.3°C, consistent with the known challenges of representing daytime temperatures over high and complex terrain. Statistical post-processing methods reduce these errors to MAE values of about 2.3°C–2.6°C, whilst the MLP clearly outperforms the other approaches, achieving MAE near 1.3°C and RMSE around 1.7°C–1.8°C. Seasonal variations in error are weak, indicating relatively stable performance throughout the year.

For TMIN (Figure 7d–f), observed temperatures range from roughly 14.3°C in winter to 16.3°C in summer, consistent with strong altitude control. ERA5 shows a systematic cold

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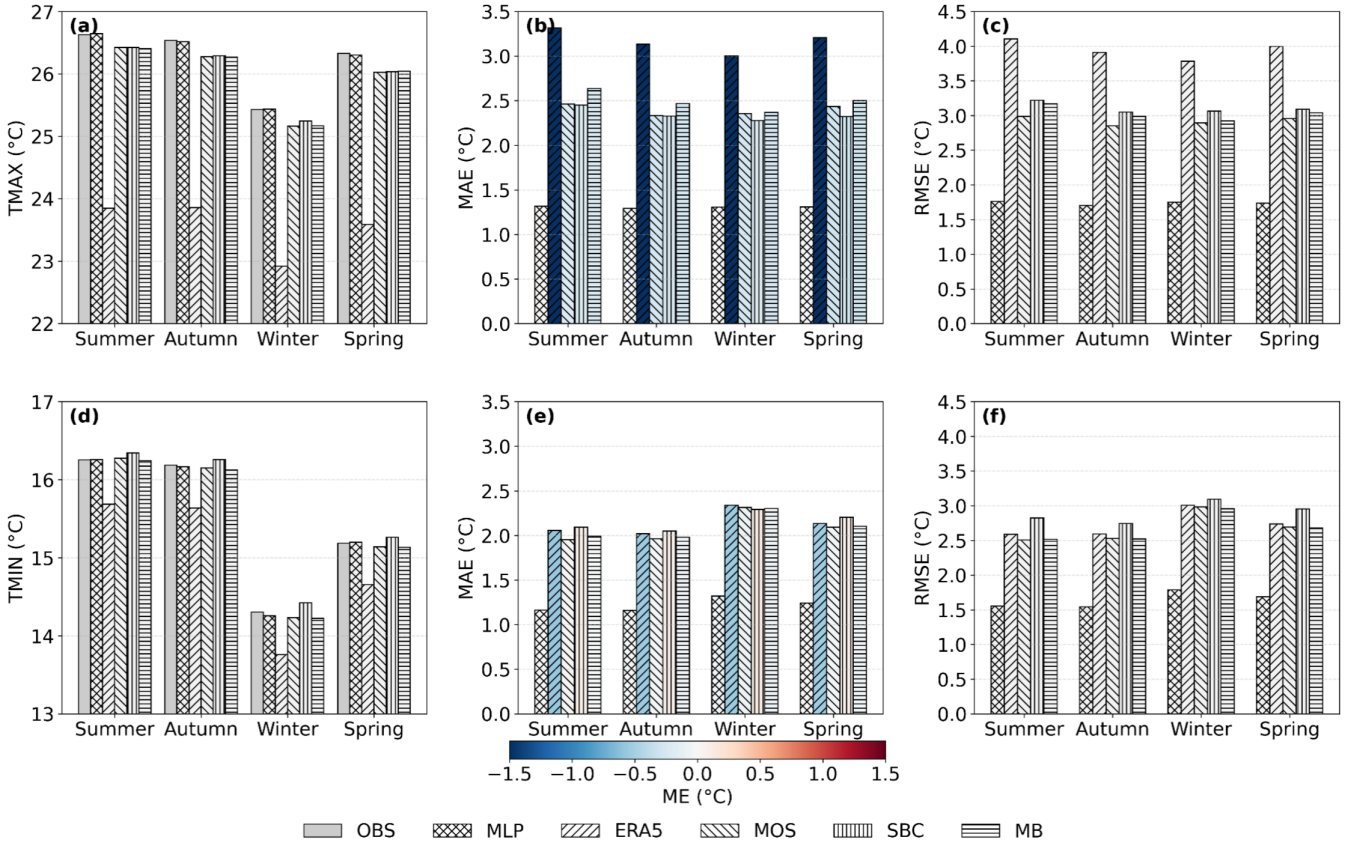


FIGURE 7 | Seasonal evaluation of MAX and TMIN for Cluster 2. Panels (a, d) show seasonal mean temperatures from observations (OBS), ERA5, and bias-corrected ERA5 using MOS, SBC, MB, and MLP. Panels (b, e) and (c, f) display the MAE and RMSE, respectively. Bar colours indicate the sign and magnitude of the ME. [Colour figure can be viewed at wileyonlinelibrary.com]

bias of about −0.5°C to −0.6°C, with MAE between 2.0°C and 2.3°C. Bias-corrected products provide moderate improvements, whereas the MLP again yields the lowest errors, reducing MAE to approximately 1.2°C–1.3°C and RMSE to about 1.6°C–1.8°C, with particularly good performance during winter.

Overall, Figure 7 highlights the strong impact of Andean topography on temperature representation in Cluster 2. Error statistics are consistently larger for TMAX than for TMIN, which is consistent with the greater sensitivity of daytime heating to elevation effects, surface processes, and complex terrain. Although bias-correction methods improve both variables, the MLP provides the most accurate and robust performance across all seasons, demonstrating its clear advantage for temperature modelling in this Andean-dominated climatic regime.

3.4 | Cluster 3: Southeastern South America (Subtropical Region)

Cluster 3 (Figure 2b shaded in purple) is located over the subtropical interior of south-eastern South America, encompassing northern Argentina, Paraguay, Uruguay and south-western Brazil. The region is characterised by low relief, predominantly below 500 m, and a humid subtropical climate influenced by warm and moist air masses from the South Atlantic and frequent frontal systems. These conditions favour strong land–atmosphere coupling, relatively high temperatures year-round, hot summers, mild winters and a pronounced seasonal cycle.

Figure 8 summarises the seasonal behaviour of TMAX and TMIN in Cluster 3 and the corresponding error statistics for the different correction methods. For TMAX (Figure 8a–c), observations show a clear seasonal cycle, with maxima in summer (~30°C) and minima in winter (~22°C). ERA5 systematically underestimates TMAX in all seasons (ME between about −0.7°C and −1.1°C), whilst MOS and MB tend to overestimate, particularly during summer and spring. Bias-correction methods substantially reduce these errors, with SBC and especially MLP yielding near-zero mean errors and reduced dispersion. The MLP provides the best overall performance, with MAE around 1.15°C–1.32°C and RMSE close to 1.7°C–1.8°C, compared to ERA5 RMSE values exceeding 2.0°C–2.3°C.

For TMIN (Figure 8d–f), errors are generally smaller than for TMAX, indicating a better representation of nocturnal conditions. ERA5 exhibits a consistent warm bias in all seasons (ME ≈ +0.6°C to +0.9°C), whereas MOS shows a cold bias, most pronounced in winter (ME ≈ −1.1°C). SBC and MLP effectively remove most of the bias, with MLP achieving mean errors close to zero year-round. MAE values range from approximately 1.0°C in summer to 1.5°C in winter, with RMSE between ~1.3°C and 2.0°C.

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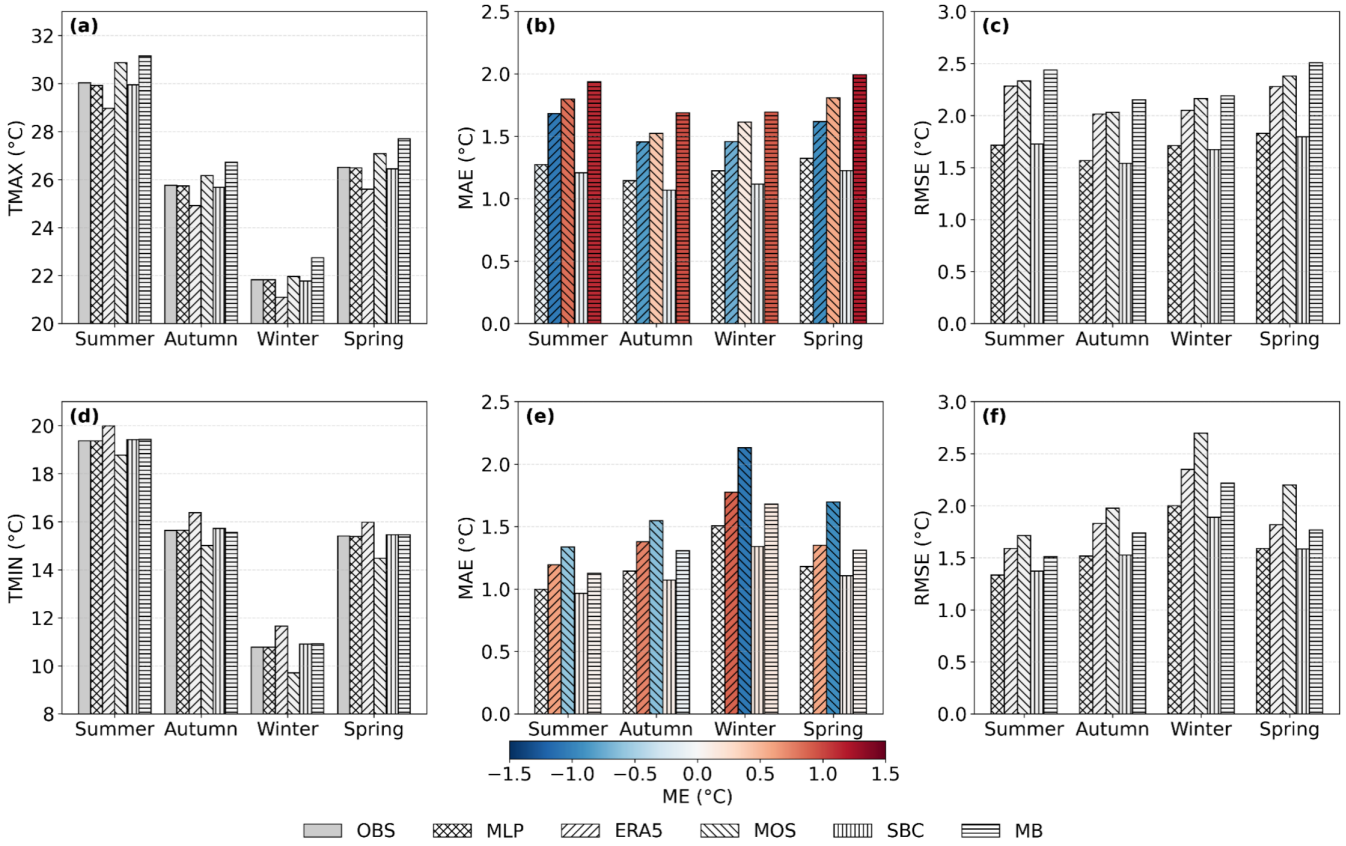


FIGURE 8 | Seasonal evaluation of MAX and TMIN for Cluster 3. Panels (a, d) show seasonal mean temperatures from observations (OBS), ERA5, and bias-corrected ERA5 using MOS, SBC, MB, and MLP. Panels (b, e) and (c, f) display the MAE and RMSE, respectively. Bar colours indicate the sign and magnitude of the ME. [Colour figure can be viewed at wileyonlinelibrary.com]

Overall, the results indicate that Cluster 3 benefits from relatively good baseline performance, consistent with its flat terrain and humid subtropical climate. Nevertheless, TMAX remains more challenging than TMIN, particularly in summer, which is consistent with the stronger sensitivity of daytime temperatures to surface radiation and land–surface processes. Across all seasons and for both variables, the MLP consistently delivers the most accurate and stable results.

3.5 | Cluster 4: Eastern Brazil (Tropical Atlantic Region)

Cluster 4 (Figure 2b shaded in pink) encompasses eastern and north-eastern Brazil, extending from the semi-arid interior to the Atlantic coastal belt, and is characterised by predominantly low elevations, including gentle plateaus and coastal plains. The regional climate is persistently warm, with high temperatures throughout the year and limited seasonal amplitude, particularly for minimum temperatures, reflecting the combined influence of tropical and subtropical air masses, strong solar forcing, and maritime moderation along the coast.

Figure 9 summarises the seasonal behaviour of TMAX and TMIN and their associated errors for this cluster. Observed TMAX exceeds 31°C in summer and spring and displays only modest seasonal variability (Figure 9a). ERA5 exhibits a systematic cold bias in TMAX (ME ≈ −0.7°C to −1.0°C), whilst MOS shows a tendency towards overestimation, most notably during winter. Statistical post-processing methods substantially reduce these errors, with MLP and SBC yielding MAE values close to 1.0°C–1.2°C (Figure 9b) and RMSE around 1.3°C–1.5°C, indicating an improved representation of daytime heating in this lowland, warm environment (Figure 9c).

In contrast, TMIN shows a clearer seasonal cycle, with winter minima near 16.5°C and summer values around 21°C (Figure 9d). ERA5 presents a persistent warm bias in TMIN, particularly during winter (ME ≈ +1.19°C), which may be related to difficulties in simulating nocturnal cooling under humid and weakly stratified boundary-layer conditions. Amongst the correction approaches, MLP consistently achieves the lowest MAE (≈1.0°C–1.4°C) and reduced RMSE across seasons (Figure 9e,f). Overall, the observed error patterns in Cluster 4 are consistent with the influence of daytime radiative and land–surface processes for TMAX, whilst TMIN biases may also be affected by boundary-layer dynamics and moisture conditions, with data-driven corrections providing the most consistent improvements for both variables.

3.6 | Sensitivity of the MLP-Based Bias-Correction to Temporally Independent Validation

In spatial bias-correction studies, the evaluation of model performance may be influenced by spatial dependence amongst observational stations. When training and validation datasets

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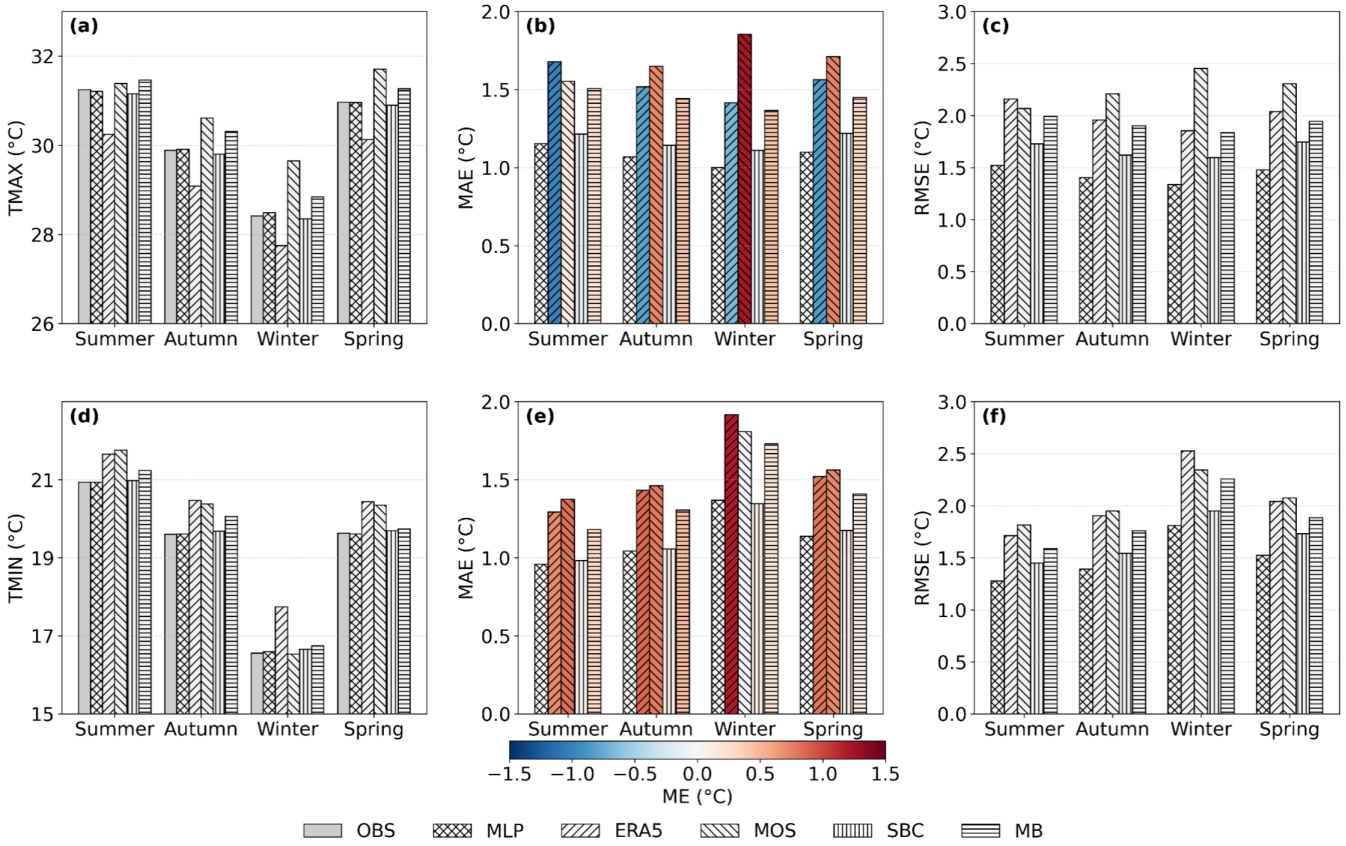


FIGURE 9 | Seasonal evaluation of MAX and TMIN for Cluster 4. Panels (a, d) show seasonal mean temperatures from observations (OBS), ERA5, and bias-corrected ERA5 using MOS, SBC, MB, and MLP. Panels (b, e) and (c, f) display the MAE and RMSE, respectively. Bar colours indicate the sign and magnitude of the ME. [Colour figure can be viewed at wileyonlinelibrary.com]

are generated through random sampling, geographically proximate stations may be included in both subsets, potentially leading to an overestimation of model skill. This issue is particularly relevant for methods such as the multilayer perceptron (MLP) and the simple bias correction (SBC) approach, whose formulations explicitly or implicitly incorporate spatial information. In such cases, the models may partially learn location-specific patterns rather than fully generalisable relationships, which can artificially enhance their apparent predictive performance.

Amongst the evaluated methods, the MLP consistently exhibited the best overall performance in terms of error reduction for both Tmax and Tmin. Whilst this result highlights the potential of the MLP for bias correction, it also raises the question of whether part of this improved performance could be influenced by spatial proximity effects within the evaluation framework. Therefore, an additional sensitivity analysis was conducted to assess the robustness of the MLP performance under a temporally independent validation strategy.

To this end, an alternative experiment was designed in which the MLP was trained using data from 2010 to 2018, and subsequently applied to correct ERA5 temperature data for the year 2019. In this configuration, the model has no prior exposure to the target year, ensuring a fully independent temporal evaluation. Furthermore, the assessment was restricted to the same subset of stations (10%) used in the original validation procedure, allowing a direct comparison between the standard random-sampling approach and the temporally independent framework.

The results indicate that the performance of the MLP remains largely consistent across both experimental setups for most clusters. For TMAX (Table 1), differences in root mean square error (RMSE) between the original experiment and the temporally independent validation are generally very small, typically within 0.01°C–0.10°C. A similar behaviour is observed for TMIN (Table 2), with only slightly larger deviations in some cases, but still within a limited range. In all clusters, the MLP substantially outperforms the original ERA5 data, and the relative differences between the two validation strategies remain minor.

An exception is observed in Cluster 2, where the differences between the two experimental setups are more pronounced. This behaviour can be attributed to the complex topography and strong spatial heterogeneity of the region, where temperature is highly dependent on local factors such as elevation, slope, and small-scale circulation processes. In this context, the random spatial split may benefit from the presence of nearby stations in the training dataset, allowing the model to better represent local relationships. In contrast, the temporally independent validation requires the model to generalise across different atmospheric conditions without relying on temporal analogues, which can lead to slightly larger errors.

These findings suggest that, although spatial dependence may contribute to a modest overestimation of model skill in specific cases such as Cluster 2, it does not fundamentally alter the overall conclusions of this study. The superior performance of the MLP is primarily associated with its ability to capture the underlying

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TABLE 1 | RMSE (°C) for TMAX across clusters.

Cluster	ERA5	MLP (random split)	MLP (temporal validation)	Temporal-random
0	2.0677	1.4260	1.4394	+0.0134
1	3.2716	2.2642	2.2572	−0.0070
2	3.8311	1.8862	1.9884	+0.1022
3	2.0314	1.6688	1.6846	+0.0158
4	1.9865	1.4114	1.4378	+0.0264

Note: ERA5 denotes the original data. MLP (random split) refers to the original evaluation (90% training, 10% validation), whilst MLP (temporal validation) uses training data from 2010 to 2018 and testing in 2019. Values correspond to the mean RMSE computed over the full year of 2019.

TABLE 2 | RMSE (°C) for TMIN across clusters.

Cluster	ERA5	MLP (random split)	MLP (temporal validation)	Temporal-random
0	1.6922	1.2560	1.2684	+0.0124
1	2.5593	1.9895	1.9951	+0.0056
2	2.8844	1.9420	2.0770	+0.1350
3	1.9346	1.6118	1.6142	+0.0024
4	2.1658	1.6006	1.6137	+0.0131

Note: ERA5 denotes the original data. MLP (random split) refers to the original evaluation (90% training, 10% validation), whilst MLP (temporal validation) uses training data from 2010 to 2018 and testing in 2019. Values correspond to the mean RMSE computed over the full year of 2019.

relationships between thermodynamic and geographic predictors, rather than being predominantly driven by spatial proximity amongst stations. Nonetheless, it is acknowledged that, for the SBC method, it was not possible to perform an equivalent sensitivity analysis within the same framework, and therefore its potential sensitivity to spatial dependence could not be fully assessed.

Overall, the additional experiment provides further support for the robustness of the MLP-based bias-correction approach, reinforcing the reliability of the results presented in this study. The consistency of the results under temporally independent validation indicates that the method is suitable not only for correcting biases in analyses and reanalyses, but also for application in predictive contexts. However, in regions characterised by complex terrain and sparse observational networks, such as Cluster 2, some caution is warranted, as model performance may be influenced by the interaction between training strategy and local spatial variability.

3.7 | Spatial Distribution of Temperature Fields and Biases

Beyond the cluster-based evaluation, the spatial analysis of temperature fields and associated biases provides important insight into the geographical coherence and physical consistency of the bias-correction approach. As the MLP method consistently outperformed the alternative techniques across all clusters and seasons, only the MLP-corrected product is considered here. Figures 10 and 11 show the seasonal spatial distributions of TMAX and TMIN over South America derived from the original ERA5 reanalysis (ERA5ori), the MLP-corrected (MLPcorr) dataset, and their differences (ERA5ori − MLPcorr).

In addition, to directly assess the effectiveness of the bias correction relative to observations, Figure 12 presents the spatial distribution of mean biases computed at station locations for both ERA5 and the MLP-corrected dataset over the period 2010–2019. This complementary analysis allows the evaluation of the magnitude and spatial structure of the original ERA5 biases and the residual biases after correction.

The original ERA5 fields successfully reproduce the large-scale climatological temperature gradients across the continent, characterised by warmer conditions over tropical lowlands and progressively cooler temperatures towards higher latitudes and elevated terrain. Despite this overall realism, systematic and spatially coherent biases are evident for both temperature variables. For TMAX (Figure 10), predominantly negative values in the difference fields indicate a widespread cold bias in ERA5, with a systematic underestimation of daytime maximum temperatures, particularly over southern South America and regions of complex topography. In contrast, TMIN (Figure 11) exhibits predominantly positive differences, revealing a pervasive warm bias associated with an overestimation of nocturnal minimum temperatures across large portions of the continent, especially in southern and subtropical regions.

The application of the MLP bias-correction method substantially attenuates these large-scale and regionally coherent biases whilst preserving the fundamental spatial structure of the ERA5 temperature fields. The corrected product systematically increases TMAX and decreases TMIN in areas where ERA5 exhibits known systematic errors, yielding seasonal adjustments typically ranging between 1°C and 4°C, consistent with the magnitude of biases identified in the station-based and cluster-level evaluations. Importantly, the ERA5ori − MLPcorr difference fields do not exhibit excessively strong or spatially abrupt corrections. This behaviour reflects the design objective of the MLP, which is trained to estimate and remove the systematic, conditionally averaged bias as a function of spatial location, elevation and seasonality, rather than to modify the underlying synoptic-scale or climatological structure of the reanalysis. As a result, the corrections remain spatially smooth, aligned with major climatic gradients and physiographic features, and free of artificial discontinuities that would indicate overfitting or physically implausible adjustments.

The relatively moderate magnitude of the spatial differences should therefore not be interpreted as a limitation of the correction. Instead, it reflects a physically consistent bias-removal process. Although ERA5 temperature biases are statistically significant and climatologically relevant, their seasonal mean amplitude is generally moderate. An effective correction is therefore expected to reduce systematic errors without introducing exaggerated spatial contrasts or artificial distortions in the

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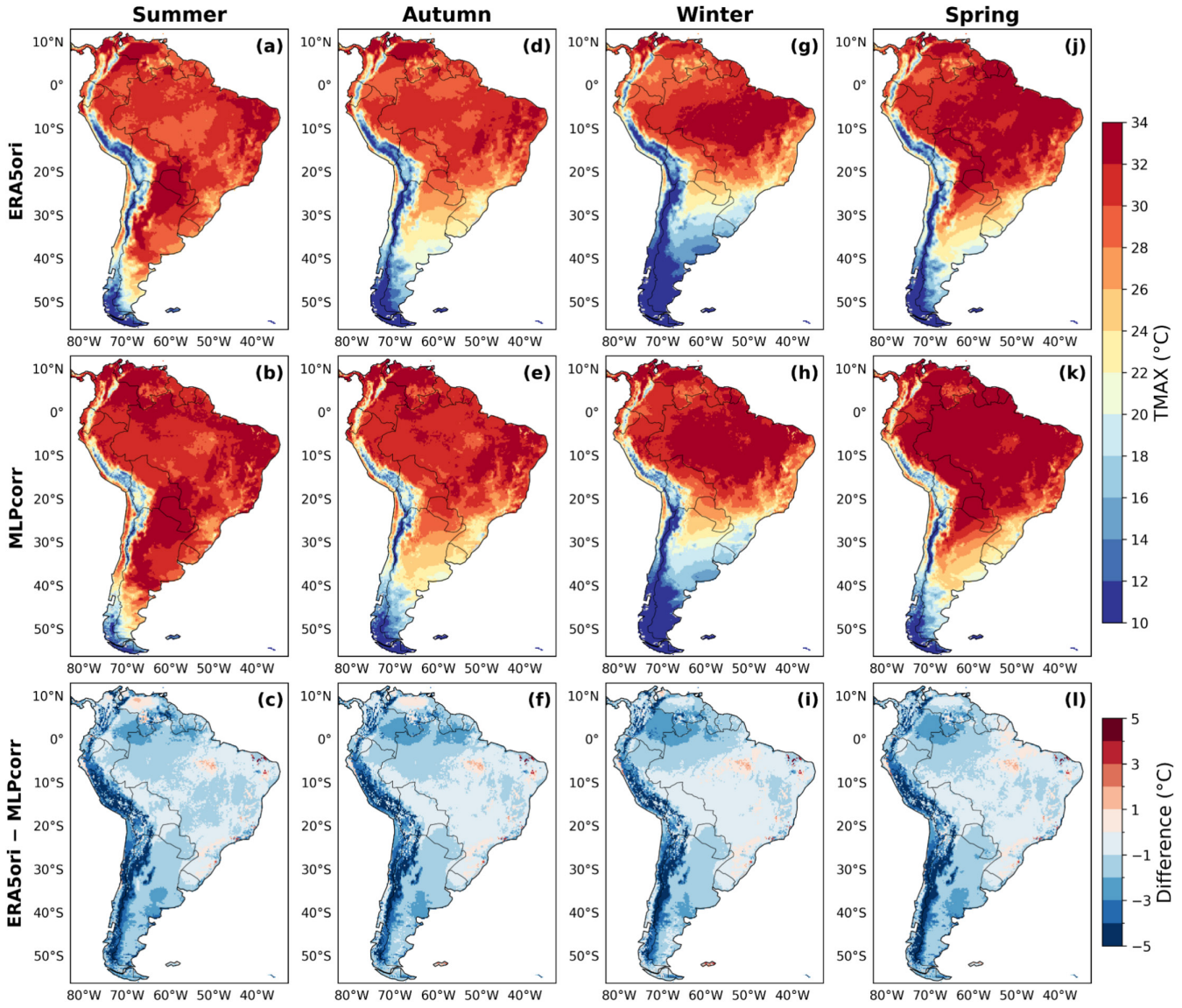


FIGURE 10 | Seasonal maps of TMAX over South America from ERA5Ori, MLP bias-corrected ERA5 (MLPCorr), and their differences (ERA5Ori − MLPCorr) for summer (a–c), autumn (d–f), winter (g–i), and spring (j–l). [Colour figure can be viewed at wileyonlinelibrary.com]

temperature fields. The spatial analysis confirms that the superior statistical performance of the MLP identified in the cluster-based assessment is also expressed at the continental scale. The corrected fields remain geographically coherent and physically plausible. The combination of bias attenuation, spatial smoothness and preservation of large-scale thermal gradients enhances the suitability of the MLP-corrected ERA5 dataset for applications requiring gridded temperature information over South America, including climate monitoring, impact assessments and model evaluation.

Figure 12 illustrates the spatial distribution of mean temperature biases for ERA5 and the residual biases after MLP correction, evaluated at station locations over the period 2010–2019. For TMAX (Figure 12a), ERA5 exhibits a widespread cold bias across most of South America, particularly pronounced over central and southern regions, with several stations showing large negative deviations. This systematic underestimation is reflected in the negative spatial mean bias ($\mu = -1.193$°C). After MLP correction (Figure 12b), the magnitude of the bias is substantially reduced, with most stations showing values closer to zero and a more balanced spatial distribution between slight under- and overestimations. The spatial mean bias is effectively removed ($\mu = -0.041$°C), indicating a strong correction of the large-scale systematic error.

For TMIN (Figure 12c), ERA5 displays an overall warm bias, especially over eastern Brazil and parts of the southern region, with several stations exceeding +2°C. This positive bias results in a spatial mean of $\mu = 0.720$°C. The MLP-corrected results (Figure 12d) show a marked reduction in both the magnitude and spatial extent of these overestimations, with most stations exhibiting near-zero residuals. The spatial mean bias is again effectively eliminated ($\mu = 0.017$°C).

Overall, the comparison highlights that the MLP approach not only reduces the magnitude of the biases but also improves their spatial consistency. Regions that previously exhibited coherent large-scale errors in ERA5 show substantially weaker and less structured residuals after correction. However, some localised

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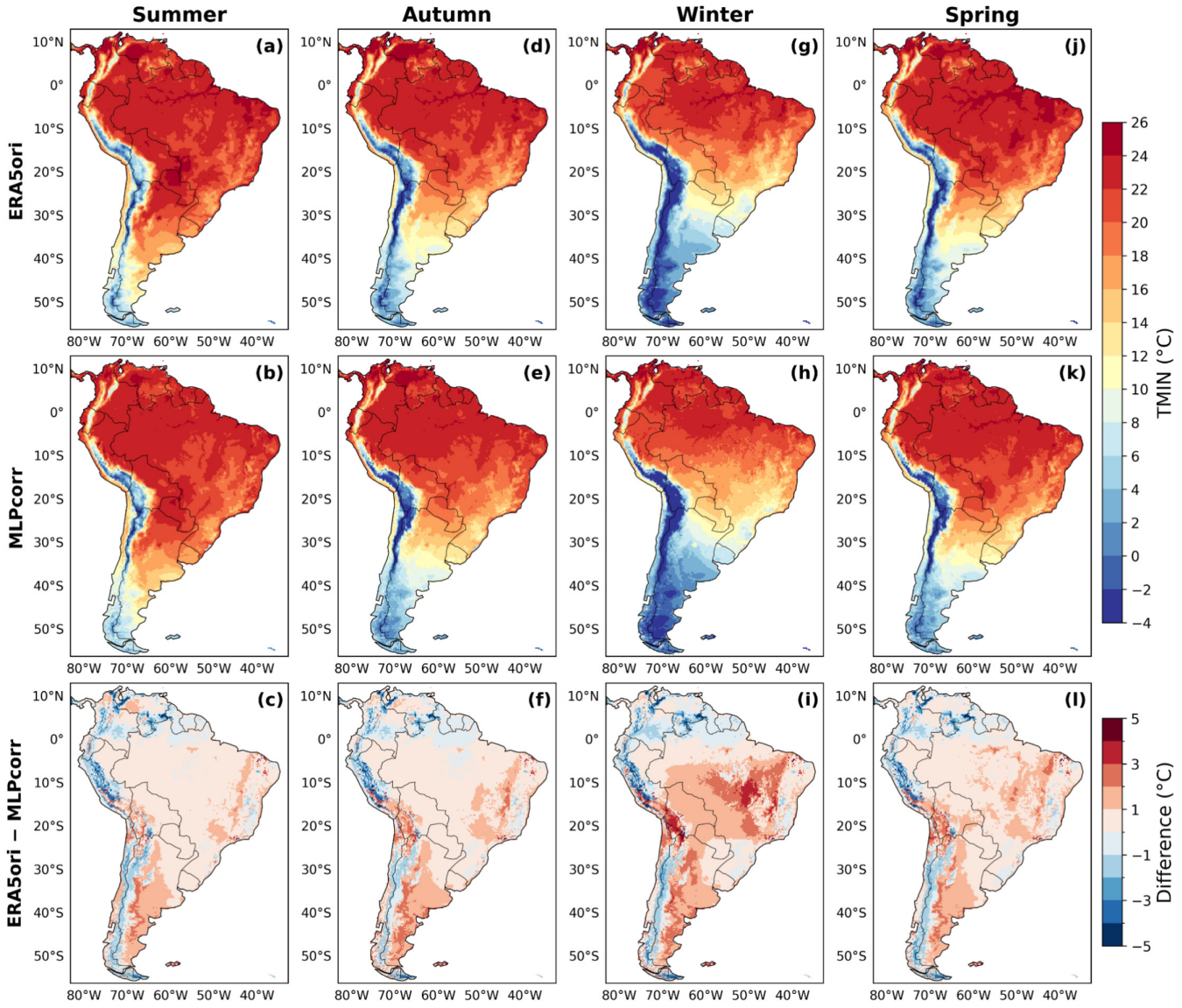


FIGURE 11 | Seasonal maps of TMIN over South America from ERA5Ori, MLP bias-corrected ERA5 (MLPCorr), and their differences (ERA5Ori – MLPCorr) for summer, autumn, winter, and spring. [Colour figure can be viewed at wileyonlinelibrary.com]

biases persist, indicating that whilst the MLP effectively mitigates systematic errors, small-scale discrepancies remain, particularly in regions with greater climatic heterogeneity or sparser observational coverage.

4 | Discussion

This study demonstrates that ERA5 near-surface temperature over South America exhibits systematic and physically consistent biases that vary with climate regime and topographic complexity. Across all analysed regions, ERA5 shows a persistent cold bias in TMAX and a warm bias in TMIN, confirming patterns previously reported for ERA5 and earlier ECMWF reanalyses (Rozante et al. 2022; Choudhury et al. 2023; Lopes et al. 2024). These biases are commonly linked to limitations in the representation of surface energy balance, cloud–radiation interactions and boundary-layer processes in reanalysis systems (Lopes et al. 2024). The underestimation of TMAX has often been associated in the literature with uncertainties in surface shortwave radiation, potentially linked to limitations in cloud representation, which may reduce daytime surface heating in reanalysis products (Du et al. 2018; Fu et al. 2026). Conversely, the overestimation of TMIN is commonly discussed as being related to deficiencies in the simulation of nocturnal stable boundary layers, where enhanced turbulent mixing and underestimated longwave radiative cooling can limit night-time temperature decreases near the surface (Sinclair et al. 2022). As a direct consequence of the opposing biases in TMAX and TMIN, ERA5 tends to underestimate the daily thermal amplitude across South America.

The largest ERA5 errors occur in mountainous regions (Clusters 2), where cold biases in TMAX exceed 2°C–3°C. These enhanced errors are primarily associated with the limited ability of coarse-resolution reanalyses to resolve steep thermal gradients (Zhao et al. 2022) and complex sub-grid topography (Sebbar et al. 2023). In such environments, unresolved elevation contrasts and valley–ridge structures affect the representation of surface heating, local circulations and lapse rates. Additionally,

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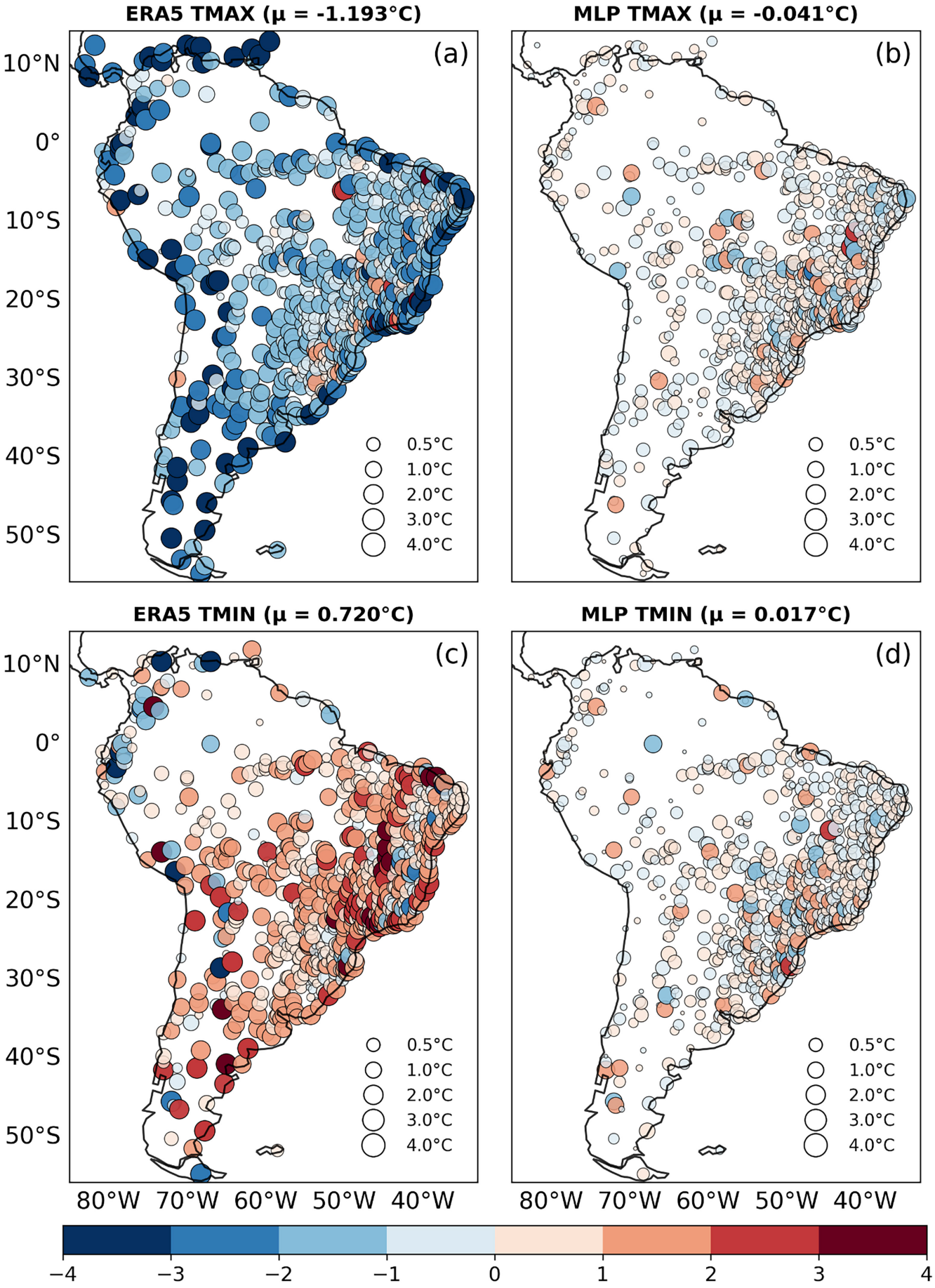


FIGURE 12 | Legend on next page.

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FIGURE 12 | Spatial distribution of mean temperature bias (°C) for (a) ERA5 TMAX, (b) MLP-corrected TMAX, (c) ERA5 TMIN, and (d) MLP-corrected TMIN over South America for the period 2010–2019. Biases were computed at station locations and averaged over time. Circle size is proportional to the absolute bias magnitude, whilst colour indicates bias sign and intensity (blue: underestimation; red: overestimation). Values in each panel represent the spatial mean bias. [Colour figure can be viewed at wileyonlinelibrary.com]

uncertainties in snow–albedo feedbacks, including the representation of snow cover and surface reflectivity, can amplify cold biases by reducing absorbed shortwave radiation during snow-covered periods (Kouki et al. 2023). These limitations are further compounded by the sparse availability of in situ observations in mountainous areas, which constrains data assimilation and increases uncertainty in near-surface temperature estimates (Bonshoms et al. 2022).

All bias-correction methods reduce ERA5 temperature errors, but their effectiveness differs substantially. Linear approaches such as Mean Bias and MOS provide partial improvements but often leave residual seasonal errors or introduce overcorrection, particularly in regions dominated by non-linear land–atmosphere interactions. In contrast, the Multi-Layer Perceptron (MLP) consistently outperforms all other methods across regions and seasons, reducing RMSE by up to 40%–45% for TMAX in the Andes and southern South America and achieving near-zero mean errors for TMIN in most clusters. Beyond these statistical gains, the spatial distribution of the corrected fields shows a clear attenuation of large-scale and topographically driven bias structures present in the original ERA5, indicating that the MLP captures regionally coherent error patterns linked to elevation, latitude, and climatic regime. These results are consistent with recent studies showing that machine-learning approaches better capture non-linear relationships between temperature, elevation, seasonality, and spatial variability than traditional statistical methods (Dhawan et al. 2024; Tanimu et al. 2024; Pasula and Subramani 2025).

The magnitude of ERA5 temperature biases varies across regions according to the dominant climatic and geographic controls, despite the consistent patterns described above. In tropical environments (Cluster 0 and parts of Cluster 4), strong cloud–radiation interactions and high atmospheric moisture likely contribute to damping the diurnal temperature range, limiting the magnitude of biases despite persistent errors in shortwave radiation and nocturnal cooling. In contrast, mountainous regions (Cluster 2) exhibit larger biases due to unresolved topographic effects, including elevation mismatches and sub-grid-scale processes, which strongly affect both radiative fluxes and near-surface temperature gradients. In extratropical regions (Clusters 1 and 3), the combined influence of synoptic variability, surface winds and boundary-layer dynamics enhances turbulent mixing, contributing to larger errors in both daytime heating and nocturnal cooling.

These results also provide a physical basis for interpreting the relative performance of the bias-correction methods. Linear approaches such as MB and MOS rely on stationary relationships and therefore have limited ability to represent regime-dependent and non-linear processes, particularly in regions where temperature is strongly controlled by cloud–radiation feedbacks, land–atmosphere coupling and complex terrain. Spatial methods such as SBC partially account for geographic variability but remain constrained by station density and interpolation uncertainties. In contrast, the superior performance of the MLP can be understood in terms of its ability to learn non-linear interactions between atmospheric variables, implicitly capturing the combined effects of humidity, radiation, elevation and seasonality. This allows the MLP to more effectively correct systematic deficiencies in the representation of surface energy balance and boundary-layer processes in ERA5, leading to more consistent improvements across all climatic regimes.

Building on the spatial and seasonal patterns discussed above, the results across all clusters consistently indicate that the MLP achieves the lowest errors for both TMAX and TMIN, demonstrating its ability to better represent complex bias structures in ERA5 temperature. A direct comparison amongst methods with different levels of complexity, however, requires careful interpretation. The MLP has greater flexibility and uses additional predictors compared to MB, MOS, and SBC, which may contribute to its improved performance. However, these predictors (latitude, longitude, elevation, and temporal encodings) are physically meaningful and largely deterministic, representing the spatial and seasonal structure of temperature rather than introducing independent meteorological information. Thus, the comparison reflects different modelling approaches: linear (MB, MOS), spatial interpolation (SBC), and non-linear spatial–temporal learning (MLP), rather than strictly equivalent configurations. In addition, all methods were calibrated within homogeneous topoclimatic clusters, ensuring a consistent climatic framework for comparison. Therefore, the results highlight the advantage of increased model flexibility in representing complex bias patterns, rather than indicating a universal superiority of the MLP.

The spatial heterogeneity of the observational network across South America plays a central role in shaping the performance of the bias-correction methods, particularly in data-sparse regions such as the Amazon Basin. In this region, the limited and uneven station coverage constrains the estimation of local statistical relationships, directly affecting the robustness of statistical methods such as MB and MOS, particularly when applied within regionally defined clusters. Because these approaches rely on regionally conditioned calibration, their performance depends on the availability of representative observations within each cluster, which is inherently limited in the Amazon. As a result, their ability to capture spatially varying and non-linear bias structures associated with tropical processes may be reduced.

In contrast, the SBC and MLP methods exhibit different sensitivities to observational sparsity. The SBC method relies on spatial interpolation of station-based corrections and is therefore strongly affected by large inter-station distances, which can lead to overly smooth correction fields and a reduced capacity to

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resolve local variability. The MLP, trained using data from all regions, is less directly dependent on local sample density and can maintain relatively strong performance through cross-regional generalisation. However, this behaviour implies that its skill in the Amazon may reflect, at least in part, the transfer of relationships learned in data-rich regions rather than a fully local representation of temperature processes. Furthermore, the limited number of stations also affects model evaluation, as validation metrics in such regions may underestimate the true spatial variability of errors.

Similar challenges arise in mountainous and southern extratropical regions, such as the Andes and southern Patagonia, where ERA5 exhibits the largest temperature biases. These errors are closely linked to the coarse spatial resolution of the reanalysis and its limited ability to represent elevation-dependent processes, complex terrain, and local circulation patterns. Although all bias-correction methods show substantial improvements in these regions, their performance remains influenced by the same observational constraints. Statistical methods such as MB and MOS, particularly when applied within regionally defined clusters, may suffer from reduced robustness due to limited local sampling, whilst SBC is affected by the large distances between stations, which degrade interpolation accuracy. The MLP, whilst benefiting from training across the full domain, may not fully capture the specific physical processes governing temperature variability in complex terrain.

Consequently, the confidence in the reported improvements in these regions is inherently lower than in data-rich areas. The combination of strong baseline biases and limited observational coverage introduces additional uncertainty in both model calibration and validation. In particular, the reduced number of stations may lead to an overestimation of model performance, as the available observations may not fully capture the spatial heterogeneity of temperature fields. Therefore, whilst bias correction significantly improves agreement with observations in mountainous and southern extratropical regions, these results should be interpreted with caution, recognising that uncertainty remains higher due to both observational limitations and unresolved physical processes.

5 | Conclusions

This study developed and evaluated a regionally informed framework for correcting systematic biases in ERA5 near-surface air temperature across South America, combining topoclimatic clustering with a range of statistical and machine-learning approaches. The regionalisation provided a physically consistent basis for comparing model performance, whilst allowing different correction strategies to account for variability at appropriate spatial scales. The results clearly demonstrate that ERA5 exhibits consistent and spatially heterogeneous biases, characterised predominantly by a cold bias in daily maximum temperature (TMAX) and a warm bias in daily minimum temperature (TMIN), with the largest discrepancies occurring in mountainous and high-latitude regions. These biases are consistent with limitations reported in the literature regarding the representation of surface energy balance and boundary-layer processes.

The application of objective clustering methods enabled the identification of five physically coherent topoclimatic regions, with the Gaussian Mixture Model providing the most consistent and interpretable partition. This regionalisation proved essential for improving the physical relevance of the analysis, as it provided a structured framework to assess model performance across distinct climatic regimes whilst accounting for spatial heterogeneity. In doing so, it helped to reduce the masking of local-scale processes and to better interpret differences amongst bias-correction approaches.

All four bias-correction methods, MB, MOS, SBC and MLP, successfully reduced ERA5 temperature errors to varying degrees. Simpler approaches such as MB and MOS provided robust and interpretable improvements, particularly in data-rich regions, although they may leave residual seasonal errors or introduce overcorrection in some cases. SBC introduced additional spatial refinement by accounting for geographical variability. The MLP, however, consistently achieved the best overall performance across regions and seasons, reflecting its ability to represent non-linear relationships and to integrate multiple physically meaningful predictors (e.g., latitude, longitude, elevation and temporal encodings). Although these predictors do not vary in time and therefore do not explicitly represent atmospheric variability, they provide essential spatial context that enables the model to better capture persistent, location-dependent biases.

Importantly, the superior performance of the MLP should not be interpreted as universal dominance, but rather as a reflection of its flexibility and ability to exploit complex spatial–temporal dependencies. Its effectiveness was especially evident in data-sparse regions, such as the Amazon Basin, where it demonstrated partial skill in generalising learned relationships from data-rich areas. A similar behaviour is observed over the Andes and other mountainous regions, where ERA5 exhibits some of the largest temperature biases due to unresolved topographic effects and complex local processes. Although all methods lead to substantial improvements in these areas, their performance remains constrained by sparse observations and the inherent limitations of the reanalysis, resulting in higher residual uncertainty. Nonetheless, the MLP shows greater robustness in these environments, likely benefiting from its ability to integrate information across regions and better represent non-linear relationships associated with complex terrain.

The results also highlight the importance of applying bias correction at the daily scale, which allows high-frequency variability associated with synoptic processes and the diurnal cycle to be more effectively represented, leading to more consistent seasonal behaviour. In addition, daytime temperatures (TMAX) remain more challenging to correct than nocturnal temperatures (TMIN), which is consistent with the stronger sensitivity of daytime temperatures to surface radiative fluxes, land–atmosphere interactions, and turbulent mixing.

Overall, the findings reinforce the importance of combining physically based regionalisation with appropriate bias-correction techniques. This approach improves the reliability of ERA5 temperature fields for climate analysis and impact studies across South America. The proposed framework enhances the

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consistency and physical relevance of the corrected data. The strong performance of the MLP, supported by the sensitivity analysis, also indicates clear potential for predictive applications. This reflects its robustness and ability to generalise across different climatic regimes. Further improvements in observational networks and continued advances in machine learning are expected to strengthen these results.

Author Contributions

Gabriela Rozante: software, visualization, writing – original draft, writing – review and editing. **José Roberto Rozante:** conceptualization, methodology, investigation, validation, formal analysis, writing – original draft, writing – review and editing.

Acknowledgements

The authors would like to express their sincere gratitude to the anonymous reviewers for their careful reading of the manuscript and for their constructive and insightful comments. Their suggestions significantly contributed to improving the clarity, robustness, and overall quality of this study. We also appreciate the time and effort dedicated to the review process. The authors also acknowledge the Georeferenced Information Base Project (BIG) of INPE, funded by the National Fund for Scientific and Technological Development (FNDCT), with financial collaboration from the Funding Authority for Studies and Projects (FINEP) and the Foundation for Science, Applications, and Space Technology (FUNCATE), "No. 01.22.0504.00."

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used in this study were obtained from the ECMWF and are publicly available through the Copernicus Climate Data Store (https://cds.climate.copernicus.eu, accessed on 19 September 2025).

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